Check for updates

Risk Analysis

WILEY

ORIGINAL ARTICLE

Reducing Risk Misinformation and Miscommunication: A Sheaf-Theoretic Perspective

Louis Anthony Cox Jr.

Cox Associates, Entanglement, University of Colorado, Denver, Colorado, USA

Correspondence: Louis Anthony Cox Jr. (tcoxdenver@aol.com)

Received: 3 June 2025 | **Revised:** 29 January 2026 | **Accepted:** 12 March 2026

Keywords: distributed risk management decision-making | misinformation | risk communication | sheaf theory | structural incoherence

ABSTRACT

Persistent misinformation poses serious challenges to risk communication, public understanding, and coordinated risk management decision-making. While past models—such as opinion dynamics, informational cascades, and the Social Amplification of Risk Framework—help explain how misinformation spreads, they do not explain when and why locally reasonable-seeming beliefs fail to cohere into a shared, accurate global understanding of risks. This paper models how beliefs are communicated, filtered, and interpreted across social and institutional networks. We draw on tools from sheaf theory—a mathematical framework for integrating local information into global structures—to diagnose and quantify situations where consistent beliefs at the local level cannot be coherently reconciled across the system. We discuss how these tools can support targeted interventions, such as trust-based bridge agents, semantic alignment, and control strategies that improve belief integration. Examples and simulations demonstrate how both classical and novel problems—such as information silos and echo chambers, fragile consensus stemming from information cascades, and agreement on words without agreement on meanings—can be reframed and addressed as design flaws in the processes used to communicate and update individual beliefs. From this perspective, risk communication should be treated not just as a matter of accurate content, but as a system design problem in which coherence, interoperability, and resilience must be deliberately engineered. We propose that tools from sheaf theory can help meet this challenge by supporting risk communication that is more likely to produce shared, accurate beliefs—and thereby enable better informed and better coordinated risk management decision-making.

1 | Introduction

In today's hyperconnected information ecosystem, false or misleading content—whether arising from misunderstanding, malice, or machine—can spread rapidly across social, institutional, and algorithmic networks, producing real-world harms that outpace correction (Jamalzadeh et al. 2022; Tambuscio and Ruffo 2019). Traditional risk-analysis frameworks often treat misinformation as a stochastic or behavioral nuisance and model its spread using epidemic or opinion-dynamics equations, suggesting that it can be corrected or limited by improving fact-checking or inoculation with warnings (Jeong and Shin 2024). Yet these useful approaches can miss a deeper, structural vulnerability: even perfectly rational, well-intentioned agents may exchange locally plausible and consistent messages that collectively form an incoherent or false global narrative, undermining effective risk understanding, preparedness, and response. From COVID-19 vaccine hesitancy and questionable folk remedies to election-related conspiracy theories and critical infrastructure disruptions, the consequences of mis/disinformation are not limited to behavioral anomalies or noise in the system; rather, they can create structural vulnerabilities in the architecture of modern risk communication, eroding trust and coordination in crises and increasing vulnerability and reducing resilience to both deliberate

© 2026 Society for Risk Analysis.

attacks and unintentional miscommunications and errors arising from unstated assumptions and divergent interpretations of risk terms (Aven and Thekdi 2022; Jamalzadeh et al. 2022).

When trust dynamics, feedback loops, or platform algorithms amplify locally plausible but globally misleading signals, the result is not merely distortion but systemic incoherence in risk understanding, response, and coordination. This dynamic has long been discussed in the Social Amplification of Risk Framework (SARF), which emphasizes how information about hazards is refracted through institutional, media, and social "amplification stations" that shape public perception and behavior (Kasperson et al. 1988, 2022). In the digital age, this process is intensified by algorithmic curation, social homophily, and viral content personalization, blurring traditional boundaries between expert, media, and lay roles. Even well-intentioned actors may unintentionally spread misinformation by trusting seemingly credible intermediaries, reinforcing polarization and feedback cycles that standard diffusion models fail to fully capture. These dynamics suggest that misinformation propagation should be understood not only probabilistically but also topologically, as exploiting structural gaps in how beliefs and information flow across interconnected yet epistemically fragmented systems.

This paper asks a new question: Can flows of misinformation in networks be detected and corrected based on the patterns of global inconsistency that they generate? To address this question, we draw on key concepts from sheaf theory, a branch of mathematics with recent applications in sensor networks, distributed decision-making, and information fusion (Rosiak 2022). At its core, sheaf theory formalizes when and how locally consistent pieces of information can be integrated (or "glued" in the metaphorical language of the theory) into a coherent whole. It provides methods to diagnose and localize structural obstructions when such integration is impossible, identifying where and why local consistency fails to scale up to global coherence.

Although sheaf theory is well established in applied topology, it has not previously been applied to risk communication or misinformation in the risk sciences. The contribution of this work is therefore not new mathematics but a translation and adaptation of sheaf-theoretic tools to a new domain, risk communication, where they provide structural diagnostics unavailable in existing models. These methods allow risk analysts to formalize when and why a shared, globally coherent, understanding of risk may be impossible, even when all local communications are honest and internally consistent. In this sense, the framework complements rather than replaces existing approaches and opens new avenues for research, intervention design, and empirical testing.

Finally, most of the examples in this paper are intentionally simple and didactic. They are not offered as realistic models of complex risk-communication systems, but as illustrations of how sheaf-theoretic concepts can be applied—much as early work on Bayesian networks or causal graphs relied on small examples to make new tools accessible before they were deployed at scale. The aim is expository: to introduce risk analysts and practitioners to a framework and set of diagnostic tools that translate qualitative notions—such as semantic drift and partial overlap of beliefs—into explicit structural tests for risk-communication analysis.

Readers with strong mathematics backgrounds, particularly in algebraic topology, can find the underlying theory developed more succinctly, precisely, and generally in existing references (e.g., Aktas et al. 2024; Curry 2013; Hansen and Ghrist 2019; Rosiak 2022). In contrast, this paper focuses on accessible applications and implications specifically for mis/disinformation in risk analysis. To keep the paper accessible and self-contained, Section 2 surveys established models of opinion dynamics; readers familiar with this material may skip directly to Section 3 on key ideas of sheaf theory. The sheaf-theoretic perspective developed from Section 3 onward is not intended as an alternative model of opinion dynamics, but as a way to pose and address new questions—for example, why even truthful sharing of risk perceptions can fail to produce a coherent shared understanding of risk, and how targeted interventions in risk-communication architectures can restore the possibility of convergence to an accurate shared understanding.

We use the generic term "beliefs" to refer to information communicated in a social or organizational network, with nodes representing agents who hold, send, and receive beliefs. The beliefs an agent communicates may differ from its private beliefs—for example, due to strategic or social distortion. We position this work alongside the SARF (Kasperson et al. 2022), traditional bounded confidence (Deffuant et al. 2000; Hegselmann and Krause 2002) and cascade models (Bikhchandani et al. 1992; Banerjee 1992) of opinion dynamics, and recent algebraic approaches to influence-network interventions. We explore several new perspectives enabled by sheaf theory, including

1. Formulation of structural misinformation as a failure of compositionality in networked beliefs.
2. Diagnostic algorithms for detecting and localizing structural misinformation "obstructions" (via cohomology calculations).
3. Prescriptive strategies for correcting misinformation flows, grounded in network structure and trust mappings.

The following sections review quantitative models of opinion dynamics and misinformation (Section 2); introduce sheaf concepts tailored to risk communication (Section 3); present expository and practical examples involving cascades, polarized clusters, and strategic filtering, together with methods for managing misinformation risk (Section 4); and conclude with implications for risk analysts and directions for future research.

2 | Background: Foundational Models of Misinformation Spread and Amplification

Misinformation and disinformation do not merely distort individual beliefs; they propagate through complex social and institutional systems, often producing disproportionate public responses, trust breakdowns, or harmful collective actions (Jamalzadeh et al. 2022). To understand and manage these risks, researchers have proposed a range of models across disciplines, including psychology, sociology, network science, and risk analysis. This section reviews representative approaches to modeling how misinformation spreads and why it persists. It also highlights

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the limitations of these models that motivate the sheaf-theoretic perspective in Section 3.

2.1 | Informational Cascades and Fragile Consensus

In classical models of informational cascades (Bikhchandani et al. 1992; Banerjee 1992), individuals update their beliefs and actions based not only on private signals but also on the observed choices of others. Early actions serve as signals about the beliefs of the early actors that can shape the inferences and subsequent beliefs and actions of those who choose later, even if they have independent private information. In these models, individuals observing the publicly visible choices of others may rationally suppress their own information if they believe that the early choices reflect better information. This creates a feedback loop in which conformity spreads, not because individuals are irrational, but because it is rational to defer to others when their choices appear to reflect better information.

Once an early pattern emerges—regardless of its validity—subsequent actors may rationally suppress their own evidence and conform. In Banerjee's (1992) sequential model, each individual chooses based on a private signal and the observed decisions of predecessors, but cannot see others' private information. After just a few consistent early actions, this limited observability leads later agents to infer that earlier decisions probably reflect high-quality signals, leading them to ignore their own information even if it contradicts the observed trend. As this pattern repeats, the system stops learning. Subsequent choices are driven by imitation rather than information. What appears to be consensus is often merely informational lock-in. Bikhchandani et al. (1992) emphasize that such cascades are typically founded on a narrow base of information—often the private signals of just the first few actors. Once a cascade begins, it ceases to incorporate new information, because later decisions reveal nothing about private information. This makes cascades highly fragile in the sense that small changes in early choices can have outsized and persistent effects on the eventual consensus. That consensus may be based on little more than inference from an initial small, unrepresentative sample. Cascades are thus a central mechanism for understanding how idiosyncratic or misleading signals can arise and propagate through otherwise rational decision systems, producing widespread but unreliable beliefs.

One application of the information cascade model provides is as a lens for examining early public messaging and risk communication during the COVID-19 pandemic. In the United States, some prominent public health and news authorities initially downplayed the effectiveness of face masks. In February 2020, Surgeon General Jerome Adams tweeted "Seriously people—STOP BUYING MASKS! They are NOT effective in preventing general public from catching #Coronavirus" (Allassan 2020). Similarly, in early March 2020, Dr. Anthony Fauci stated that masks might make people feel better but were not providing the protection people thought they were. CBS reported that "When it comes to preventing coronavirus, public health officials have been clear: Healthy people do not need to wear a face mask to protect themselves from COVID-19. 'There's no reason to be walking around with a mask,' infectious disease expert Dr. Anthony Fauci told 60 Minutes" (McCandless Farmer 2020). These statements were motivated in part by a broader effort to conserve masks for healthcare workers amid shortages.

However, such messaging contributed to public skepticism and confusion about mask usage, even after guidelines changed in April of 2020 to recommend masks for the general public. Despite later evidence and reversals in guidance, many individuals continued to dismiss mask effectiveness—partly due to informational cascades built on early, trusted statements. Private beliefs were suppressed or reframed to conform with perceived public consensus. Many individuals conformed to early messaging—even after their private judgments shifted—creating an informational lock-in driven more by the structure of belief propagation than by continued misinformation.

2.2 | Simple Quantitative Models of Opinion Dynamics

Relatively simple quantitative models of *opinion dynamics*, in which each agent's beliefs adjust over time to reflect the beliefs of others with whom the agent communicates, have been developed extensively over the past half century. In such models, beliefs are typically represented by numbers, such as the probabilities that agents give to a proposition being true. Adjustments are typically modeled by systems of differential or difference equations describing how beliefs are updated. Even such simple models can yield rich dynamics that suggest, and perhaps illuminate, aspects of real-world opinion dynamics such as convergence to a shared opinion (DeGroot 1974) or failure to converge and emergence of polarized views (Friedken-Johnson 1990).

Table 1 provides a comparative overview of several foundational models in opinion dynamics, highlighting how each framework captures different aspects of belief evolution within social networks.

Each row in Table 1 highlights a dimension along which different models capture distinct features of belief evolution in networks. These features are briefly explained next before turning to a more detailed discussion of each model.

- **Belief update rule**: Describes how agents revise their opinions over time. Most classical models assume averaging, either across all neighbors (DeGroot), or across all those whose expressed beliefs fall within a confidence window (Hegselmann–Krause [HK], Deffuant–Weisbuch [DW]); or combine such averaging with a model of resistance to change in opinions (Friedkin–Johnsen).
- **Fixed network weights**: Indicates whether agents' influences on each other remains constant over time or adapt dynamically based on opinion proximity.
- **Anchoring to initial belief**: Reflects whether agents remain partially committed to their original views, even after repeated social exposure.
- **Bounded confidence**: Represents the idea that agents ignore others whose beliefs are too dissimilar—an abstraction of cognitive filtering or trust thresholds.

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Comparison of opinion-dynamics models.

Feature	DeGroot (1974)	Friedkin and Johnsen (1990/1999)	Hegselmann and Krause (2002)	Deffuant et al. (2000)
Belief update rule	Linear averaging of all neighbors	Linear averaging + weighted anchoring to initial belief	Averaging over agents within confidence bounds	Pairwise averaging if within confidence bound
Fixed network weights	Yes	Yes	No (dynamically adjusts based on confidence interval)	No (interaction only occurs between close agents)
Anchoring to initial belief	No	Yes	No	No
Bounded confidence	No	No	Yes (confidence threshold ε)	Yes (confidence threshold ε)
Consensus guaranteed?	Yes (under mild conditions)	No (opinions may stabilize without consensus)	Not guaranteed—may yield multiple clusters	Not guaranteed—multiple clusters likely
Belief heterogeneity allowed?	Temporarily, but vanishes	Persistently (via anchoring term)	Persistently (via confidence gaps)	Persistently (via confidence gaps)
Accounts for misinformation?	No	Partially (indirectly via anchoring)	No	No
Strategic distortion modeled?	No	No	No	No
Empirical realism	Limited (too simple for polarized or fragmented networks)	Better fit to persistent disagreement in real social settings	Captures polarization and fragmentation	Captures polarization and convergence patterns

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

- **Consensus guaranteed?**: Indicates whether the model mathematically ensures eventual convergence to a common opinion across all agents.
- **Belief heterogeneity allowed?**: Highlights whether final belief distributions can reflect persistent diversity or polarization.
- **Accounts for misinformation?**: Notes whether the model explicitly accommodates false, misleading, or distorted inputs.
- **Strategic distortion modeled?**: Captures whether agents deliberately misreport beliefs, as might occur under political or reputational incentives.
- **Empirical realism**: Qualitatively evaluates how well the model captures observed dynamics in real-world networks, including polarization, fragmentation, and resilience to correction.

Each model is discussed next in terms of these features.

2.2.1 | DeGroot Model (1974)

The DeGroot model is a foundational framework in opinion dynamics that describes how individuals in a network update their beliefs through social interactions. In this model, each agent holds an initial belief about a particular subject. Over time, agents revise their beliefs by averaging the opinions of their neighbors, with the influence of each neighbor determined by a predefined trust weight. These trust weights are organized into a matrix where each row sums to one, ensuring that the updated beliefs remain within a valid range.

In the classical DeGroot model, agents update their beliefs by taking a fixed linear average of their neighbors' beliefs. The social network is encoded as a stochastic weight matrix W, with entry w_{ij} indicating how much influence agent j has on agent i's opinion. A key feature of the DeGroot model is its prediction of consensus formation. Under certain conditions—specifically, when the network is strongly connected and aperiodic—the model guarantees that all agents will eventually converge to a common belief. This consensus is a weighted average of the initial beliefs, where the weights reflect the relative influence of each agent within the network. Agents with higher influence have a greater impact on the final consensus.

The DeGroot model ensures convergence to a consensus belief under strong connectivity and aperiodicity, but it does not guarantee convergence to a correct or well-calibrated belief. In particular, if the network gives disproportionate influence to a poorly informed or extreme agent, the consensus may reflect misinformation rather than truth. This concern is formalized in a theoretical extension by Acemoglu et al. (2010), who show that the presence of "forceful" agents—those who influence others but remain impervious to influence themselves—can prevent efficient information aggregation and lead to systematically biased group beliefs, even when most agents are well-informed. However, in large networks, if each agent begins with an initial belief that is an independent, unbiased estimate of the true state (e.g., a noisy signal of a scalar quantity, with bounded variance, which is automatically satisfied when beliefs are probabilities in [0, 1]), then the consensus belief under DeGroot dynamics converges to the true state if and only if no single agent or small group retains disproportionate (i.e., asymptotically nonzero) influence as the network grows (Golub and Jackson 2010). Otherwise, belief convergence may be biased or systematically incorrect—especially when influential agents are misinformed or initial beliefs are correlated (ibid).

While the DeGroot model offers a clear mathematical depiction of opinion convergence, it has limitations in capturing more complex social phenomena. For instance, it does not account for scenarios where agents might disregard opinions that differ significantly from their own (bounded confidence), nor does it model strategic behavior where agents might intentionally misrepresent their beliefs. Additionally, the model assumes static trust weights and does not incorporate mechanisms for misinformation or the persistence of belief diversity. In short, despite its elegance and simplicity, the DeGroot model is limited by its key assumptions:

- Static trust weights, rather than adaptive social influence;
- Full communication, ignoring bounded confidence effects;
- No misinformation, assuming inputs are truthful;
- No strategic behavior, overlooking the possibility that agents may deliberately misreport;
- Inevitable consensus, failing to capture polarization or persistent disagreement.

These limitations motivated the development of richer models, discussed next, that allow for selective attention, partial resistance to social influence, and greater belief heterogeneity. Nonetheless, even the more sophisticated opinion-dynamics models may fall short in addressing more structurally complex distortions such as partial communication, semantic ambiguity, or coordinated misinformation, for which sheaf-theoretic tools offer structural diagnostics and potential remedies.

2.2.2 | Example: Misinformation Drift in a DeGroot Opinion-Dynamics Model

This example illustrates how DeGroot-style averaging can converge smoothly to a shared belief even when that belief is structurally distorted by a persistent, uncorrected signal. To illustrate the behavior of the classical DeGroot model under conditions that include a persistent misinformation source, we simulate a network of five agents, each with an initial belief about the level of risk posed by a hypothetical hazard (e.g., a chemical or vaccine). The agents represent different stakeholder types: a government regulator, a corporation, two citizens, and an activist group. Beliefs are real numbers between 0 and 1, interpreted as subjective probabilities or levels of perceived risk. The *initial beliefs* are as follows:

Regulator: 0.20; Corporation: 0.25; Citizen A: 0.20; Citizen B: 0.30; Activist: 1.00

The activist thus starts with a dramatically inflated risk belief relative to the other agents, simulating a misinformation node

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

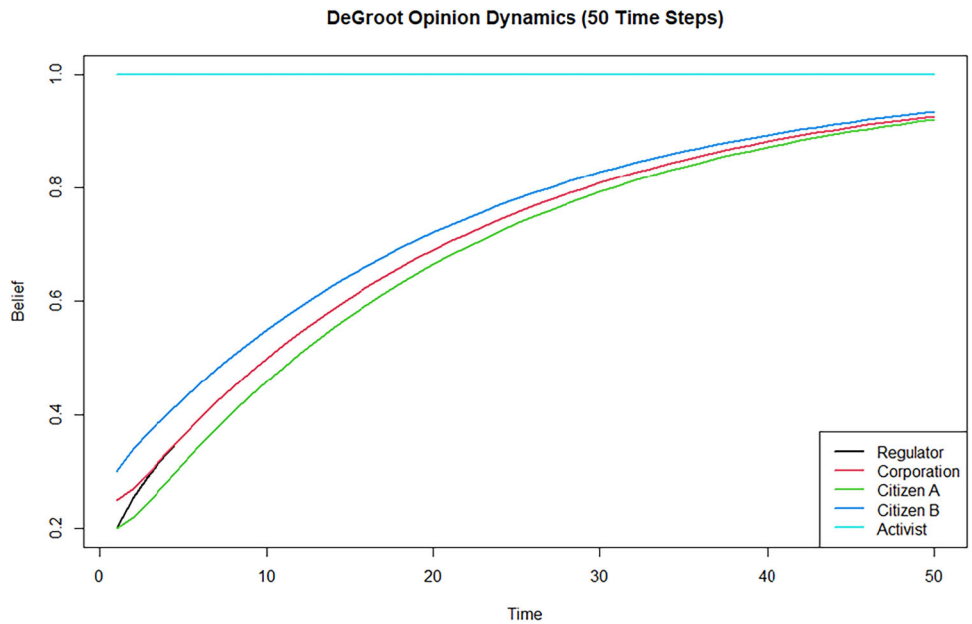

FIGURE 1 | Belief convergence in a five-agent DeGroot model with a persistent misinformation source. *Source:* Appendix A contains the R code for the DeGroot model and Figure 1.

(one possible interpretation, although this simple model does not address the objective correctness of beliefs or consider any evidence other than the beliefs themselves). In the DeGroot framework, each agent updates its belief by taking a weighted average of the beliefs of others in the network. These trust weights are represented as a row-stochastic (i.e., each row sums to 1) matrix W, where entry w_{ij} is the weight that agent i assigns to the opinion of agent j. The matrix used is: $W =$

[0.60, 0.20, 0.10, 0.05, 0.05], # Regulator
[0.30, 0.50, 0.10, 0.05, 0.05], # Corporation
[0.20, 0.20, 0.50, 0.10, 0.00], # Citizen A
[0.10, 0.20, 0.10, 0.50, 0.10], # Citizen B
[0.00, 0.00, 0.00, 0.00, 1.00], # Activist (self-trusting only)

The fifth agent (activist) assigns full weight to its own belief and zero weight to others, making it a nonupdating, persistent signaler. Other agents assign only 5%–10% weight to the activist, reflecting limited but nonzero exposure to the misinformation signal.

Figure 1 shows the evolution of beliefs over 50 time steps using the simple linear dynamics described by $x(t+1) = Wx(t)$, where $x(t)$ is the vector of agent beliefs at time step t and $x(0) = (0.20, 0.25, 0.20, 0.30, 1.00)$. Despite the activist's low influence weights, its extreme and persistent belief exerts a persistent pull on the others. All nonactivist agents ultimately converge toward a shared belief close to 0.93, far from their initial values near 0.25. This illustrates a form of misinformation drift: the DeGroot model faithfully computes local weighted averages, but fails to recognize that the global belief pattern is being distorted by a single, uncorrected extreme source. The model has no mechanism for detecting or resisting incoherent signals or testing global consistency. From a structural perspective, local agreement masks global inconsistency. This highlights a key motivation for the sheaf-theoretic methods introduced in Section 3: to diagnose when locally plausible beliefs do not compose into a globally consistent understanding, and to identify where interventions (e.g., trust reweighting or filtering) may be most effective.

2.2.3 | Friedkin–Johnsen Model (1990/1999)

The Friedkin–Johnsen model extends the classical DeGroot framework by incorporating anchoring to initial beliefs, thereby allowing agents to retain a degree of independence or "stubbornness" in the face of social influence. This modification reflects the idea that individuals may adjust their views based on what others believe, but do not completely abandon their original convictions—even after repeated exposure to alternative viewpoints. This example illustrates how partial anchoring to initial beliefs can prevent convergence to a shared belief even when communication is honest and the network is connected.

Let $x(t)$ denote the vector of agent beliefs at time t, and $x(0)$ the vector of initial beliefs. Let W be a row-stochastic influence matrix (as in the DeGroot model), and let A be a diagonal matrix of anchoring weights, where the diagonal entry $A_{ii} = \alpha_i$ represents agent i's strength of attachment to the initial belief $x_i(0)$. The update rule is:

$$x(t+1) = Ax(0) + (I - A)Wx(t)$$

This means each agent updates by blending two components: a fixed commitment to their original beliefs, $Ax(0)$; and a socially influenced component, $(I - A)Wx(t)$. If all agents have the same anchoring strength alpha (i.e., $A = \alpha I$), the rule simplifies to:

$$x(t+1) = ax(0) + (1 - a)Wx(t)$$

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

When $\alpha = 0$, the model reduces to the DeGroot model, in which agents fully adopt social input. When $\alpha = 1$, agents are completely stubborn and never update. For values in between, agents partially adjust while remaining anchored to their starting views.

This structure allows the model to produce persistent disagreement: the system often stabilizes without reaching consensus. Final beliefs reflect a weighted compromise between initial positions and social influence. This dynamic captures key features of real-world belief systems, such as those seen in public debates over vaccines, climate change, or chemical safety, where deeply held prior beliefs persist despite repeated messaging from authorities or peers. The Friedkin–Johnsen model also allows for exploration of the effects of individual stubbornness (via the anchoring matrix A), influence centrality (how the network structure affects long-run beliefs), and long-term diversity of opinion in the presence of misinformation or partial resistance to correction. For these reasons, the model is particularly relevant for risk analysts studying belief polarization, partial misinformation uptake, and the limits of consensus-building in public health and environmental communication.

To illustrate how anchoring to initial beliefs can prevent convergence and sustain persistent disagreement, we simulate belief updating in the Friedkin–Johnsen model using the same five-agent network as in Figure 1. Appendix A contains the R code for the simulation. As before, agents represent different stakeholder types: a government regulator, a corporation, two citizens, and an activist group. Each agent begins with a real-valued belief between 0 and 1 representing a subjective probability or level of perceived risk associated with a hypothetical hazard. Initial beliefs are set as follows:

$x(0)$: Regulator: 0.20; Corporation: 0.25; Citizen A: 0.20; Citizen B: 0.30; Activist: 1.00.

The same row-stochastic influence matrix W is used as in Figure 1 to capture relative trust among agents. For this simulation, the anchoring strengths entry $A_{ii} = \alpha_i$ representing agent i's are set to:

Regulator: 0.1; Corporation: 0.2; Citizen A: 0.3; Citizen B: 0.2; Activist: 1.0.

This configuration means that the activist is fully stubborn—assigning zero weight to social input—while the other agents blend their initial beliefs with social influence to varying degrees. The belief update rule $x(t+1) = Ax(0) + (I - A)Wx(t)$ ensures that each agent remains partially anchored to its starting point throughout the process.

Figure 2 shows the evolution of beliefs over 50 time steps. Unlike in the DeGroot model, the agents do not converge to a shared belief. Instead, their trajectories stabilize at different equilibrium values that reflect a compromise between original convictions and social influence. The activist remains fixed at a belief of 1.0, exerting upward pressure on others but never adjusting. The regulator and corporation gradually move toward intermediate values, while Citizen A (with a higher anchoring weight) ends up with the lowest final belief.

This simulation illustrates how persistent disagreement can emerge endogenously from partial anchoring, even in a strongly connected network. In contrast to the DeGroot model, where local averaging masks global distortion, the Friedkin–Johnsen model captures the limits of belief convergence in the presence of initial conviction. For risk analysts, this dynamic reflects the challenge of overcoming entrenched positions in domains such as vaccine safety, environmental hazards, or controversial regulations. Even when accurate information is widely shared, partial resistance to influence can prevent full alignment, allowing polarized or skewed belief states to persist indefinitely.

2.2.4 | Hegselmann–Krause Model (2002): Clustering of Beliefs Under Bounded Confidence

The Hegselmann–Krause (HK) model modifies classical opinion dynamics by introducing a *bounded confidence threshold*: agents only consider the beliefs of others whose beliefs fall within a certain distance—denoted by epsilon, ϵ—of their own. In other words, they update their beliefs based only on those whose opinions are not too different from their own This reflects a simple but psychologically realistic principle: individuals are generally unwilling to engage with views that differ too greatly from their own, effectively filtering out extreme or distant opinions. The following example illustrates how bounded confidence can produce stable clusters of local agreement while preventing system-wide belief convergence.

In the HK model, each agent holds a continuous-valued belief (e.g., a real number between 0 and 1) and updates it by averaging the beliefs of others who fall within its confidence interval. Formally, the belief of agent i at time $t+1$ is:

$$x_i(t+1) = \text{average of } x_j(t) \text{ for all } j \text{ such that } |x_j(t) - x_i(t)| \leq \epsilon$$

That is, $x_i(t+1)$ is computed by averaging the beliefs of all agents whose beliefs are within ϵ of $x_i(t)$. All agents update synchronously. Influence is symmetric: if agent i considers j within its confidence bound, then j also considers i, provided their beliefs remain within epsilon of each other.

Because the set of influencing agents in the HK model changes dynamically based on belief distances, the model exhibits nonlinear and path-dependent behavior, unlike the fixed-weight, linear updating in the DeGroot and Friedkin–Johnsen models. This nonlinearity gives rise to several distinct emergent phenomena. Depending on the value of the bounded confidence threshold (epsilon) and the initial distribution of beliefs, the population may converge to complete consensus (if epsilon is large and initial beliefs are already tightly clustered). For smaller values of epsilon (i.e., less open-minded agents), it may instead fragment into multiple stable belief clusters (a form of polarization), or may become stuck in frozen deadlock if subgroups move outside each other's confidence bounds and become mutually unresponsive. The eventual outcome is highly sensitive to initial conditions: small changes in either the distribution of initial beliefs or the value of epsilon can shift the long-run behavior from agreement to persistent fragmentation. Thus, relatively minor variations in trust, framing, or background knowledge can substantially affect group coherence.

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

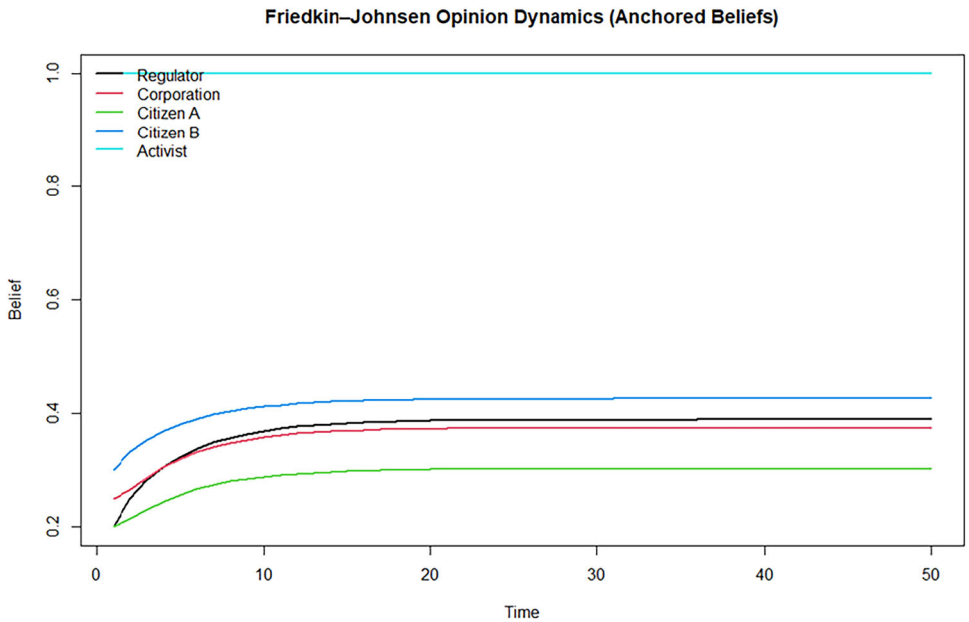

FIGURE 2 | Anchored beliefs in a Friedkin–Johnsen opinion-dynamics model. *Source:* Appendix A contains the R code for the Friedkin–Johnsen model and Figure 2.

The HK model also captures an important psychological feature of opinion dynamics: selective exposure. Agents are not strategic or deceptive in this model, but they selectively update only from others whose beliefs are already similar enough to feel cognitively comfortable. The model assumes honest communication and symmetric influence, but it allows for the endogenous emergence of informational silos. These features make the HK model particularly relevant for understanding echo chambers, filter bubbles, and the persistence of ideological divides in domains such as public health, environmental policy, and risk regulation. The bounded confidence mechanism provides a simple yet powerful account of why some individuals may discount or ignore messages that deviate too far from their current beliefs. As a result, even widely disseminated expert consensus can fail to influence skeptical or polarized subgroups.

Fragmentation and disagreement can thus persist—not because communication channels are absent, but because belief receptivity is inherently bounded. Like the Friedkin–Johnsen model, the HK framework suggests that persistent belief heterogeneity may be a structural consequence of cognitive and informational constraints, rather than a breakdown in messaging or rationality. It shows that local agreement can coexist with system-wide disagreement, especially in settings where belief updating depends on psychological proximity as much as informational accuracy.

To illustrate how bounded confidence shapes belief dynamics, we simulate the HK model with 10 agents, each assigned a random initial belief between 0 and 1. The bounded confidence threshold, epsilon, is set to 0.2. At each time step, an agent averages the beliefs of all agents whose beliefs fall within epsilon of its own. Figure 3 shows the evolution of beliefs over 10 time steps, starting from independently and uniformly drawn initial values. Despite all agents operating under identical update rules and honest averaging, the bounded confidence mechanism fragments the population into distinct belief clusters. These clusters emerge rapidly and persist, as agents outside each other's confidence bounds cease to influence one another. Arguably, this dynamic captures key qualitative features of real-world opinion landscapes, including the emergence of echo chambers and opinion polarization. Even in the absence of misinformation or strategic distortion, the simple cognitive filter of bounded confidence can lead to long-term disagreement. Agents with initially extreme beliefs may become isolated, while more moderate agents coalesce into one or more central clusters.

For risk analysts and communicators, this model illustrates why even factually accurate, widely shared messages may fail to generate consensus: if the initial belief gaps are too wide, informational isolation and self-reinforcing subgroups can arise naturally. This has practical implications for understanding public responses to health advisories, environmental risk communication, and emerging technologies. The HK model shows that disagreement may persist not because of bad information, but because of bounded receptivity to alternative views.

Further insight into the mechanisms underlying belief fragmentation and polarization is offered by Milli (2021), who develops a generalized bounded confidence opinion-dynamics model to study the formation of polarized belief clusters under varying social conditions. Extending the HK framework, Milli incorporates heterogeneity in agent trust levels, degrees of open-mindedness, and susceptibility to social influence. The model reveals that polarization can persist or even intensify when agents interact preferentially with those they trust or perceive as similar, and when a subset of agents remains epistemically rigid. Crucially, the results show that even well-calibrated, nonmanipulative agents can evolve into opposing camps when the network topology enforces homophily or inhibits bridging interactions. This reinforces the view that fragmentation and misinformation susceptibility may not arise from irrationality or bad-faith actors alone, but from structural properties of social networks and

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

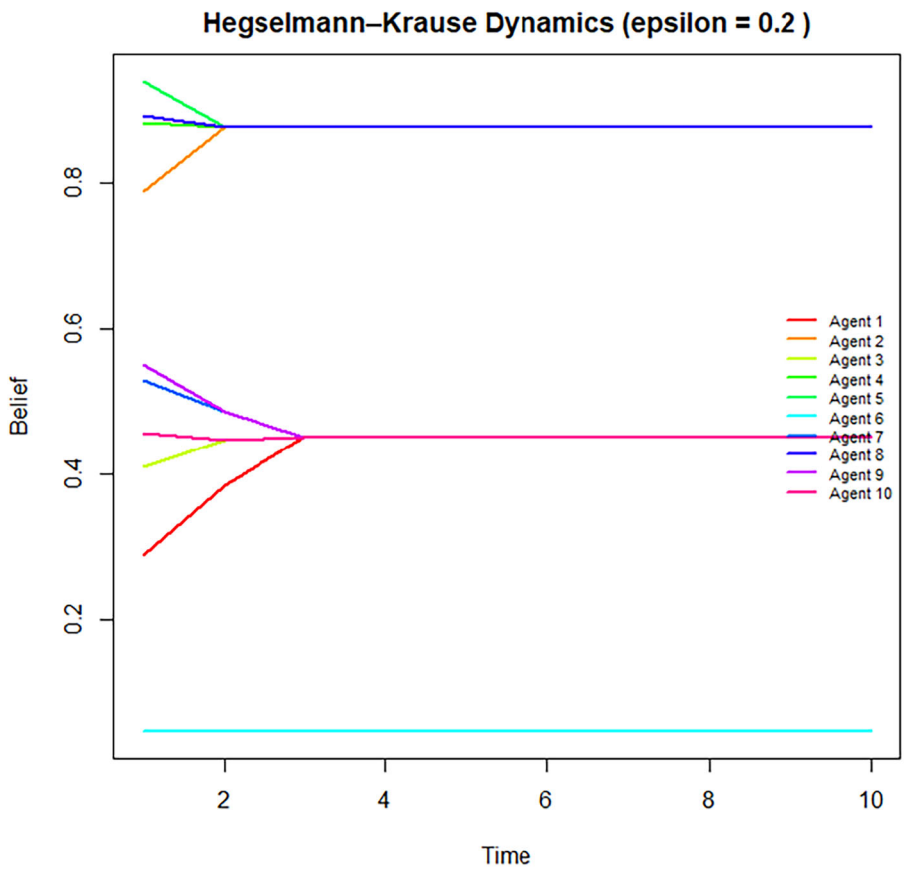

FIGURE 3 | Clustering under bounded confidence in the Hegselmann–Krause model. *Source:* Appendix A contains the R code for the Hegselmann–Krause model and Figure 3.

bounded confidence rules. For risk analysts, this highlights the importance of identifying not only belief outliers or extreme messages, but also latent trust asymmetries and topological bottlenecks that prevent information from diffusing coherently across groups.

Recent extensions of the HK model explore the effects of *manipulative agents*—agents whose opinions follow a predefined trajectory and are immune to influence from others but can affect normal agents within their confidence bounds. Bautista (2025) shows that even a relatively small group of such agents can systematically shift the opinions of the rest of the population over time, effectively dragging consensus or central clusters toward more extreme positions. This mechanism offers a quantitative account of Overton window dynamics—the gradual normalization of previously fringe beliefs—by modeling how incremental shifts from persistent, coordinated message sources can realign the bounds of socially acceptable opinion. Analytical results show that the manipulators' influence depends on their speed of opinion change, their number relative to normal agents, and the confidence threshold ϵ. Slow, coordinated manipulative shifts can cause substantial and lasting opinion realignment even without necessarily involving misinformation or deception.

Wu and Guo (2024) introduce a framework that incorporates both online and offline interactions, as well as agents with varying epistemic behaviors. In their model, agents interact with online neighbors whose opinions differ from their own by no more than a certain confidence level (ϵ), and with offline neighbors in a small-world network, regardless of opinion differences. They also introduce two types of epistemically irresponsible agents: those who deliberately disseminate misinformation and those who uncritically follow the majority opinion. Simulations reveal that broader confidence intervals can make agents more susceptible to misinformation, especially when interacting primarily online. However, incorporating offline interactions and Bayesian updating mechanisms can mitigate the spread of misinformation, highlighting the importance of diverse information sources and critical thinking in opinion dynamics.

2.2.5 | Deffuant–Weisbuch Model (2000)

This example illustrates how decentralized, pairwise belief updating can lead to fragmented local convergence without global agreement. The DW model is a bounded confidence opinion-dynamics framework that, like the HK model, assumes that individuals only engage with others whose beliefs are sufficiently similar. However, it differs from HK in a key structural feature: rather than updating synchronously by averaging across all nearby beliefs, the DW model uses asynchronous, pairwise interactions. At each time step, a pair of agents is selected at random, and they update their beliefs toward each other—but only if the difference between their current beliefs is less than a predefined confidence threshold, epsilon (ε). If the two agents' opinions are within epsilon of each other, they move closer by

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

a fixed proportion μ toward the average of their current beliefs. Formally, if agents i and j with beliefs x_i and x_j interact, and $|x_i - x_j| < \varepsilon$, then both update according to:

$$x_i(t+1) = x_i(t) + \mu\left(x_j(t) - x_i(t)\right)$$

$$x_j(t+1) = x_j(t) + \mu\left(x_i(t) - x_j(t)\right)$$

Typical values for μ are between 0 and 0.5, with $\mu = 0.5$ corresponding to full averaging between compatible agents. If the belief gap exceeds ε, no update occurs, and the agents retain their current opinions.

This asynchronous, pairwise update mechanism introduces important qualitative differences from the HK model. Because only two agents update at a time, belief changes are slower and more path-dependent, and the system's evolution depends more strongly on the sequence of interactions. Unlike the HK model, where clusters form almost deterministically from initial conditions, the DW model exhibits stochastic emergence of opinion clusters over time, with outcomes varying across simulation runs. As in the HK framework, bounded confidence plays a central role in generating fragmentation and polarization. If ε is small relative to the initial opinion spread, then multiple noninteracting subgroups may emerge. If ε is large enough, the population may converge to consensus—but the process will be slower and more sensitive to interaction history than in HK or DeGroot-style models.

The DW model captures the psychological realism of selective exposure: agents only engage with others they already perceive as sufficiently similar. It also models dyadic trust thresholds—agents must be sufficiently aligned to influence each other. Unlike the Friedkin–Johnsen model, however, the DW framework lacks mechanisms for anchoring to initial beliefs; once agents begin updating, they may drift arbitrarily far from their starting points if their interactions permit it. This makes the DW model particularly useful for simulating the slow, contingent diffusion of opinions in large populations, especially when communication is sparse or decentralized. For example, in public risk debates or social media environments, people often interact one-on-one or in small groups, forming judgments through repeated local encounters. The model shows how such structures can yield fragmented convergence: local agreement within groups, but persistent disagreement between them.

For risk analysts, the DW model provides insight into how opinion drift, cluster hardening, or polarized deadlock can emerge even without misinformation or strategic behavior. It highlights the importance of interaction structure and threshold effects in shaping long-term belief landscapes. In particular, it suggests that interventions aimed at building consensus (e.g., public messaging campaigns) may fail if bounded confidence constraints prevent critical cross-group interactions from occurring in the first place.

Figure 4 illustrates the dynamics of the DW for 10 agents whose beliefs evolve through random pairwise interactions. Similar to the HK model in Figure 3, initial beliefs are independently drawn from a uniform distribution over the interval [0, 1], reflecting a wide range of perspectives or initial uncertainty about a shared issue. At each step, a pair of agents is randomly selected to interact. If the absolute difference between their beliefs is less than or equal to a fixed confidence threshold (epsilon = 0.2), both agents adjust their beliefs partially toward one another using a symmetric update rule with influence rate $\mu = 0.5$. Otherwise, the pair ignores each other and retains their current beliefs. Unlike the HK model—which applies synchronous, global updates—the localized, stochastic pairwise updates in the DW model result in slower diffusion and greater path dependence. In this run, two distinct clusters of opinion form. One group of agents stabilizes around a lower belief value near 0.2, while another cluster settles in the upper range above 0.7. No further convergence occurs once intergroup distances exceed the confidence threshold. The final configuration *exhibits fragmented convergence*: local consensus within each cluster but persistent divergence between them. Because of the random update sequence, different runs with the same parameters may yield different clustering patterns.

The DW model shows that decentralized, bounded interaction rules can produce lasting belief diversity, even in fully connected populations. For risk analysts, this dynamic is relevant to settings in which individuals form beliefs through repeated interpersonal exchanges—such as informal conversations, social media interactions, or peer deliberation—rather than broadcast-style communication. The model underscores how structural features like limited tolerance for disagreement (bounded confidence) and interaction asymmetry (pairwise exchange) can yield persistent polarization even without misinformation, anchoring, or intentional deception. As in the HK model, failure to reach consensus may reflect the natural limits of interpersonal influence rather than a failure of rationality or communication.

While classical bounded confidence models such as HK and DW help explain belief clustering and polarization, they assume static social networks and symmetric, attraction-only influence. Del Vicario et al. (2017) extend these models to better reflect real-world confirmation bias and online social behavior. Their Unbounded Confidence Model (UCM) and Rewired UCM (RUCM) allow agents with sharply divergent views to repel each other or sever ties, mimicking the dynamics of echo chambers and homophily. These modified models produce stable polarization—the coexistence of two enduring opinion groups—without requiring exogenous misinformation or anchoring. This supports the view that observed fragmentation may emerge endogenously from the structure of interactions and belief-updating rules, and provides a natural link to the notion, explored later, of structural inconsistency.

In parallel to the continuous-valued bounded confidence tradition exemplified by the preceding models, Butts et al. (2023) build on a different modeling approach known as binary-state opinion dynamics. In binary-state models, each agent holds one of two possible opinions—such as belief in true information versus belief in disinformation—and updates this belief based on interactions with neighboring agents. This framework contrasts with continuous models in which beliefs can take on a range of values; instead, binary models capture categorical decisions (e.g., "accept" or "reject") and are well suited for studying tipping-point dynamics and policy interventions. Using this framework, Butts et al. simulate disinformation spread in networks that include a small fraction of *committed agents*—individuals who persis-

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

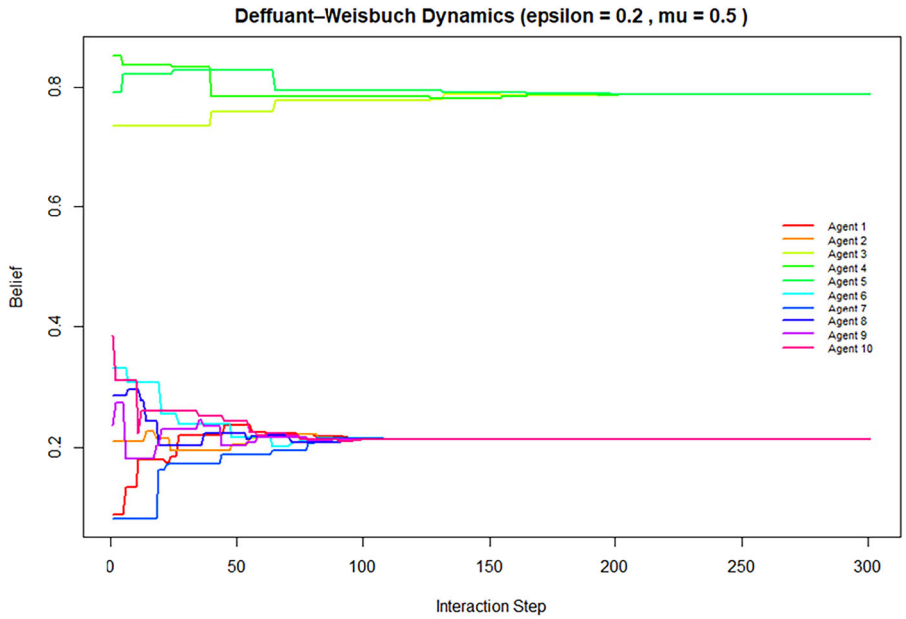

FIGURE 4 | Fragmented convergence in the Deffuant–Weisbuch model. *Source:* Appendix A contains the R code for the Deffuant–Weisbuch model and Figure 4.

tently promote false beliefs and never change their minds. They evaluate the effects of several real-world mitigation strategies, including content moderation, countermessaging campaigns, and general skepticism education (which reduces agents' likelihood of adopting any belief uncritically). Their findings, based on simulation experiments in synthetic network environments, show that increasing skepticism is particularly effective at raising the threshold needed for disinformation to dominate, while targeted interventions like moderation or countercampaigns are only effective under favorable structural conditions. These results underscore that belief outcomes depend not only on the quality of information but also on the network architecture and the rules by which agents revise their beliefs.

While classical opinion-dynamics models provide useful theoretical insights, real-world misinformation propagation often involves multiplatform interactions, heterogeneous agent behaviors, and adaptive responses to content moderation. A recent agent-based study by Murdock et al. (2024) models cross-platform information diffusion between Reddit and Twitter using empirically validated scenarios tied to three major US political events. The model includes distinct agent types—normal users, good actors, and bad actors—and platform-specific structural constraints. Their results show that moderation on one platform (e.g., Reddit) can lead bad actors to migrate to another (e.g., Twitter), where diffusion is faster and moderation may be delayed or less effective. For example, suppressing misinformation on Reddit caused a spillover effect: Twitter misinformation exposure and trust increased due to displaced content and actors. Conversely, moderation on Twitter, with its more centralized structure and faster dynamics, sometimes reduced misinformation spread on Reddit as well. These findings underscore key limitations of classical models that assume fixed topologies, static trust, and platform isolation. Real-world misinformation exploits structural differences, feedback loops, and cross-network inconsistencies.

Despite their diversity, all of the models reviewed in this section share some major limitations for modeling misinformation and disinformation. Perhaps most conspicuously, they treat beliefs as numbers and belief updates as smooth, quantitative processes, assuming sincere, boundedly rational agents who revise beliefs based on numerical proximity. In contrast, real-world misinformation often exploits structural, semantic, and contextual features of information flow—such as who is trusted, which content is salient, how messages are framed or distorted and how they are interpreted. As pointed out by Aven and Thekdi (2022), misinformation can arise from use of unclear definitions and repeated rounds of simplification, selection, and emphasis as results are transmitted down chains, such as from scientists to regulators to the press to activists to the public. Second, the models presented so far lack a framework to model structural inconsistency—that is, situations where beliefs appear locally rational or justified (e.g., pairwise agreement between sender and receiver) but cannot be consistently integrated across the network. This is where sheaf theory offers a powerful extension, enabling detection of belief patterns that cannot be coherently "glued together" across contexts, even when individual updates appear reasonable.

Misinformation and disinformation also have other important aspects that are not easily captured by simple quantitative models of opinion dynamics. Important potential sources of distortion in the production and communication of risk information that cannot necessarily be modeled well by simple quantitative models include use of terms and concepts that are not clearly defined, explained or justified; results and conclusions being reported without the assumptions on which they depend; suppression of important contributors to risk and uncertainty, such as by use of point estimates that do not reflect uncertainty and variability; motivated or agenda-driven selection of input data and assumptions; selective reporting or emphasis on which scenarios and output results to highlight; and the usual array

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of cognitive heuristics and biases, such as overconfidence and confirmation bias (Aven and Thekdi 2022). To model the influences of such nonquantitative factors on opinion dynamics, the following sections explore the use of sheaf-theoretic methods. The next section introduces key sheaf-theoretic concepts, terms, and methods—including discourse sheaves (Hansen and Ghrist 2021), restriction maps, global sections, and cohomology—to formalize and diagnose failures of belief integration in social systems. These tools allow us to move beyond scalar opinion models to structurally aware representations of misinformation flow, framing, and correction.

3 | Sheaf-Theoretic Tools for Analyzing Misinformation and Disinformation

The classical models reviewed above—such as averaging, bounded confidence, and cascade-based approaches—are valuable for analyzing how beliefs spread, converge, or fragment through repeated local interactions. However, these models are primarily designed to detect failures of convergence or disagreement at the level of *pairwise or local interactions*. They do not, in general, diagnose a distinct class of failures in which all local interactions are internally consistent, yet the system as a whole cannot support a coherent global belief assignment. In such cases, disagreement or distortion does not arise from noisy updating, strategic behavior, or irrationality at the level of individual agents, but from structural constraints on how beliefs are translated and composed across contexts. As a result, these models may report apparent local consistency or convergence even when no globally coherent interpretation of risk is possible. The sheaf-theoretic framework introduced next is designed precisely to identify and analyze these forms of structural incoherence.

3.1 | Key Sheaf Theory Concepts and Terms and Their Interpretation

This section introduces key elements of sheaf theory as they apply to misinformation systems. For orientation, a top-level view, to be refined and made more concrete momentarily, is that a *sheaf* is a mathematical structure that assigns *data* values—such as beliefs, risk estimates, or expert judgments—to *parts* of a social network or other risk-communication system (e.g., the individual agents, institutions, or shared communication forums in the system), together with rules—called *restriction maps*—for relating or comparing that data across overlapping (e.g., communicating) parts. In the context of risk analysis and misinformation, a sheaf models how information is distributed across different sources and how it is translated, simplified, or interpreted across organizational, cognitive, or other boundaries. It allows us to ask whether locally valid beliefs—such as individual assessments or partial observations that satisfy all known local constraints (i.e., are consistent across overlapping regions according to the restriction maps)—can be coherently integrated into a consistent global understanding of risk. If not, the sheaf structure helps identify precisely where and why such integration fails. This provides a powerful framework for diagnosing structural inconsistencies in expert elicitation, public communication, and multiagent decision-making.

Table 2 provides a concise reference for the sheaf-theoretic terminology used in this section; the text below develops the conceptual and mathematical relationships among these elements.

More precisely, we focus on how these elements interact in social-network settings to impose consistency constraints and to determine whether locally reasonable beliefs can be coherently integrated at the system level. *Nodes* in a network represent agents, groups, organizations, or other contexts for which beliefs are defined. Importantly, a sheaf assigns data not only to individual nodes, but also to overlapping contexts ("patches"), so that beliefs can be compared and constrained where communication occurs. These overlapping contexts serve as the domains over which beliefs are expressed, interpreted, and potentially reconciled.

Patches (also called "base patches") are the local contexts over which beliefs are organized. In a social network, they include not only individual nodes (agents) but also overlapping regions such as edges (social ties), triangles (small groups), or larger cliques (institutional alignments). Collectively, the patches form a covering of the network: a family of overlapping local regions that together span the system. In a sheaf, each patch may hold belief data, and the overlaps between patches identify exactly where beliefs must agree—after translation—if the system is to be coherent. *Edges* and higher order patches represent these shared or overlapping contexts explicitly. They function as the "glue" of the belief system, marking the points at which consistency must be enforced. In sheaf-theoretic terms, data assigned to a shared patch must be compatible with the restrictions of data from the adjacent patches it connects.

At each node (and, more generally, at each patch), a sheaf specifies a *stalk*: the space of possible beliefs, assessments, or messages that can be held in that local context. Stalk values may be scalars (e.g., a probability that a claim is true), logical propositions, belief vectors, probability distributions, or more complex structures. Stalks may differ across the network, but they are not independent: they are linked by restriction maps that impose consistency conditions across overlaps.

In this framework, a *(local) section* is a particular assignment of values from the relevant stalk to a given patch. The sheaf structure then determines whether assignments that are locally consistent on overlaps can be extended to larger regions—for example, from edges to triangles, or from dyads to cliques. This clarifies when apparently reasonable local beliefs can (or cannot) be "glued" together into broader coherent patterns. The mechanism that connects these local contexts is provided by *restriction maps*. When two nodes communicate, beliefs must be translated across a restriction map—a function describing how beliefs in one context appear in another. Such maps capture semantic mismatches (e.g., when one actor uses numeric scales and another uses categorical labels), as well as selective disclosure, framing effects, reinterpretation, simplification, exaggeration or emphasis, and other forms of distortion. In social-network sheaf models, restriction maps formalize how beliefs held at one node are expressed or perceived at adjacent nodes. Mathematically, restriction maps are deterministic functions (often linear maps in the settings emphasized below) that translate belief assignments over one patch into messages on its overlaps, thereby defining

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Key sheaf-theoretic terms used in this paper and their interpretation for risk communication.

Sheaf-theoretic term	Interpretation in risk communication
Node	An agent, group, institution, or communication context where beliefs, risk assessments, or judgments are formed or held.
Patch (base patch)	A local context over which beliefs are organized; in networks this includes individual nodes and overlapping regions such as edges, small groups, or aligned subnetworks that together cover the system.
Edge/higher order patch	An overlapping context (e.g., a social tie, small group, or institutional alignment) where beliefs or messages from different nodes must be mutually compatible.
Stalk	The space of possible beliefs, assessments, or messages an agent or institution can hold or express in a given context (e.g., scalar risks, probability distributions, categorical levels).
(Local) section	A specific assignment of beliefs or messages drawn from a stalk for a given patch; sections may sometimes be extended to larger regions if they remain mutually consistent.
Restriction map	A rule describing how beliefs in one context are translated, simplified, framed, or distorted when communicated to an overlapping context (e.g., semantic compression, selective emphasis, or reinterpretation).
Message	The expressed output of a restriction map: what an agent actually communicates to others, which may differ from its private or internal belief.
Global section	A system-wide assignment of beliefs that is compatible across all nodes, patches, and restriction maps; its existence represents a fully coherent shared understanding of risk.
Cohomology (H^1)	A diagnostic for structural incoherence: it identifies irreducible inconsistencies where all local communications appear consistent, yet no globally coherent belief assignment exists.
Sheaf Laplacian	A generalized diffusion operator that models how beliefs or risk perceptions propagate through a network over time while respecting semantic translation and local compatibility constraints.

local consistency constraints across the network. These constraints require that data assigned to overlapping regions agree when translated across those regions via the maps.

3.2 | Key Sheaf Theory Methods and Calculations

A central concept in the sheaf-theoretic framework is that of a *global section*. A global section exists when beliefs can be assigned to all nodes (and patches) in a way that is fully compatible across all overlaps and restriction maps. In this case, local belief assignments can be coherently "glued" into a single, contradiction-free, system-wide perspective. When no such assignment exists, the belief system is *structurally incoherent*: even though beliefs may be locally reasonable and mutually compatible in overlapping contexts, they cannot be reconciled into a globally consistent understanding of risk.

Sheaf cohomology provides a precise way to diagnose such failures of global coherence. Informally, cohomology identifies irreducible "holes" (unavoidable inconsistencies) in a network's ability to integrate local beliefs into a coherent whole. More formally, it enumerates and classifies the distinct ways in which local consistency fails to extend to global consistency—situations in which belief assignments agree on all overlaps, yet no globally coherent configuration satisfies all constraints simultaneously. In these cases, no global section exists. Crucially, such obstructions are not visible to pairwise agreement checks or standard consensus diagnostics; they reflect structural misalignments in how beliefs are translated and constrained across the system. Cohomology therefore distinguishes failures of integration due to system architecture from failures attributable to misinformation, irrationality, or bad faith at the level of individual agents.

In many settings relevant to risk analysis, these structural obstructions can be characterized using the *first cohomology group*, denoted H^1. In simple terms, H^1 detects *loops of inconsistency*: patterns of local belief differences that cancel out along individual overlaps but cannot be explained by any consistent global belief assignment. In linear settings, H^1 can be computed using standard matrix operations, providing a practical computational basis for applying sheaf-theoretic diagnostics to data. The following details are not essential to understand the rest of the sheaf-theoretic framework, but they provide the practical computational basis for applying cohomology to data.

Specifically, two matrices are constructed. The matrix D^0 encodes how belief domains (patches) overlap, with each row representing an overlap (such as an edge between two nodes) and each column corresponding to a patch or node; its entries capture signed differences between beliefs in overlapping regions. The matrix D^1 encodes how these overlaps themselves assemble into higher order structures, such as cycles or loops (e.g., triangles in the network). Together, D^0 and D^1 (the first and second coboundary operators in algebraic topology) describe the system's compositional structure and determine whether local belief differences can be explained by a consistent global assignment.

If a residual vector ω of local belief differences satisfies $D^1\omega = 0$ but cannot be written as D^0x for any global belief vector x, then ω represents a nontrivial cohomology class—a signal that the system's structure itself prevents gluing. In linear–algebraic terms, the number of such independent structural obstructions is

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

given by

$$\dim(H^1) = \dim(\ker(D^1)) - \dim(\mathrm{im}(D^0)).$$

When $\dim(H^1) = 0$, the system is structurally glueable and a global section exists; when it is greater than zero, the system contains irreducible inconsistencies that prevent global coherence despite local agreement.

The number of obstructions is just the number of ways that locally consistent belief differences can form loops that pass all local checks but still cannot be explained by any consistent global belief assignment—that is, the number of distinct inconsistencies in the system's structure. When $\dim(H^1) = 0$, the belief system is structurally glueable and H^1 is called trivial. When it is greater than zero ("nontrivial" cohomology), the system contains irreducible inconsistencies—much like a group of friends where every pair agrees on a version of a story, but following the story around the full circle reveals contradictions. Each local conversation seems coherent, but no single narrative can reconcile them all. Such failures of global coherence, despite local harmony, are exactly what H^1 counts. The following sections illustrate these structural failures using real and hypothetical examples of risk communication interpreted from the sheaf-theoretic perspective.

Although the exposition above emphasizes linear settings, the underlying logic applies much more broadly. Cohomology in general measures which locally compatible structures cannot be explained as boundaries of higher order structures, regardless of whether the local data are numbers, logical statements, policies, or probability distributions. What cohomology always reveals is whether a system's architecture permits consistent global reasoning from locally valid parts. For example, in a regulatory setting, stalks might contain region-specific rules, and restriction maps could enforce compatibility across shared jurisdictions; a nontrivial H^1 would reveal irreconcilable policy loops. In every case, cohomology detects structural obstructions that prevent local agreement from extending to global coherence. A further sheaf-theoretic tool, the *sheaf Laplacian*, models how information, beliefs, or risk perceptions diffuse through a network over time while respecting both structural and semantic constraints. Like the averaging of neighboring opinions in classical models of opinion dynamics, the sheaf Laplacian governs the flow of influence and belief updating. However, it is more expressive: it can operate on diverse belief types—including numerical values, probability distributions, or suitably encoded categorical expressions such as *High*, *Medium*, and *Low*—by embedding them into spaces that support linear operations. Crucially, it also incorporates the effects of restriction maps, allowing information to be translated or distorted as it moves across contexts. As a result, sheaf Laplacians are well suited for simulating how misinformation or misaligned risk signals evolve over time and for identifying stable belief patterns such as consensus, polarization, or fragmentation under network-constrained updating rules.

Building on these ideas, *discourse sheaves* (Hansen and Ghrist 2021) provide a structured model of belief and communication over social or institutional networks. In this framework, each node (e.g., a person, group, or organization) is associated with a stalk representing the space of *private* beliefs or opinions it can hold, while each edge or overlapping patch is associated with a *discourse space* that represents what can be communicated in that interaction. Nodes are linked to their neighboring patches by restriction maps that describe how private beliefs are expressed, framed, or translated in specific communicative contexts.

This structure allows agents to hold rich, high-dimensional, or private beliefs while expressing simplified, filtered, or context-dependent versions of those beliefs to others. The sheaf encodes consistency conditions at the level of expressed beliefs: agreement occurs when messages align within their shared discourse context, even if underlying private beliefs differ. In this setting, *messages* are the concrete outputs of restriction maps—that is, what an agent actually conveys about its internal beliefs to others in a given discourse space. Such messages may therefore be simplified, selectively framed, or systematically distorted representations of private beliefs, shaped by both network structure and communicative context.

In this framework, it is useful to distinguish between an agent's private beliefs and the beliefs it expresses to others. Private beliefs refer to the full, internal assessments or judgments an individual or institution may hold, which can be rich, uncertain, or high-dimensional. Expressed beliefs, by contrast, are what an agent actually communicates in a given interaction—often simplified, framed, or filtered versions of its private beliefs, shaped by context, audience, or institutional constraints. In the sheaf-theoretic model, these expressed beliefs are represented as messages, which arise as the outputs of restriction maps translating private beliefs into communicable forms in specific discourse contexts.

Taken together, the sheaf-theoretic framework—and discourse sheaves in particular—enables analysts to go beyond identifying false statements or clusters of disagreement and instead to pinpoint why locally consistent communication fails to yield coherent collective understanding. By making explicit the roles of translation, context, and structural constraints, these models reveal how miscommunication and misinformation can arise even in the absence of irrationality or bad faith.

A practical question is whether the sheaf theory approach is computationally tractable, so that it can be applied to large, real-world communication networks without excessive computation. A short answer is yes, although an exact analysis of computational complexity requires specialized methods (Koutis et al. 2011). The key computational steps, such as constructing the sheaf Laplacian and checking where messages do or do not fit together coherently, use the same kinds of sparse matrix operations and computational linear algebra and graph algorithms already well-developed in modern network analysis and data science (Hansen and Ghrist 2019). These operations scale almost linearly with the number of communication links (edges) in the network (Spielman and Teng 2011), meaning that doubling the number of links roughly doubles the computing time. Because the sheaf-Laplacian approach belongs to the same mathematical class as the well-studied network Laplacians (Hansen and Ghrist 2019), the computational burden is tractable, comparable to that of standard large-scale network analyses already familiar in applied data science.

Together, restriction maps, cohomology, sheaf Laplacians, and global sections as benchmarks provide a powerful toolkit for analyzing misinformation not just as flawed content or irrational

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

updating, but as the result of deeper structural constraints on coherence. Sheaf-theoretic modeling shifts the question from "Who misinformed whom?" to "Why can't these views be reconciled, and how might they be?" This perspective invites new forms of intervention, such as:

- Rewriting restriction maps to preserve nuance across platforms,
- Inserting bridging messages where cohomology reveals breakdowns,
- Using Laplacian dynamics to simulate how beliefs might evolve under revised constraints.

Sheaf theory thus complements behavioral, algorithmic, and network models by focusing on the compositional integrity of beliefs across the social-informational system—and how that integrity can be lost or repaired.

3.3 | Misinformation via Risk Amplification (SARF) and Semantic Ambiguity

The *Social Amplification of Risk Framework (SARF)* introduced by Kasperson et al. (1988) offers one of the earliest integrative models for how information about hazards is filtered, distorted, or amplified as it moves through media, institutions, and social networks. SARF conceptualizes these systems as "amplification stations" that do not merely relay risk messages but actively shape their salience, framing, and emotional tone—producing ripple effects on public perception, behavior, and institutional trust.

In sheaf-theoretic terms, *amplification stations*—such as media outlets, institutions, or influencers—are best understood not as restriction maps themselves, but as nodes or patches within a discourse sheaf. Each such station has an associated stalk, representing the space of messages or beliefs it can express (e.g., headlines, public statements). What connects these patches to others in the belief network are the restriction maps, which describe how beliefs held or expressed in one context (e.g., a scientific institution) are translated into those expressible or perceivable in another (e.g., a media outlet or public audience). These maps model processes like simplification, framing, selective emphasis, or distortion—capturing precisely the transformations emphasized in the SARF. From this perspective, the phenomenon of amplification corresponds to how restriction maps shape the flow of belief data across overlapping patches, altering what information is preserved, omitted, or transformed. This sheaf-theoretic framing offers a principled way to analyze when such transformations maintain coherence—and when they introduce structural inconsistencies that may propagate misinformation or obstruct shared understanding.

3.3.1 | Example: Sheaf-Theory Description of Social Amplification

This example illustrates how successive, locally reasonable semantic projections in a risk-communication chain can fail to compose into a globally coherent representation of the underlying risk. To fix ideas, we can apply the concepts of sheaf theory to model information distortion in a very simple toy example of risk communication. Suppose we have three agents in a belief network:

- S: a scientific advisory committee
- M: a media outlet
- P: the general public

Each is represented as a node (patch) in a discourse sheaf, with associated stalks:

- F(S): a high-dimensional space of technical risk assessments (e.g., probabilistic models, confidence intervals)
- F(M): a lower dimensional space of media messages (e.g., headlines or simplified summaries)
- F(P): a scalar-valued belief space (e.g., perceived danger or trust level)

The ***restriction maps*** are:

- $r_{S \to M}$: the translation from technical assessments to a media narrative. This might downweight uncertainty, emphasize worst-case outcomes, or omit qualifiers (e.g., "low-probability, high-consequence" becomes "scientists warn of risk").
- $r_{M \to P}$: the mapping from the media's narrative to public perception—further simplified or filtered through prior beliefs.

Now suppose:

- The scientific committee's actual assessment is that a certain chemical poses a *low risk with high uncertainty*.
- The media simplifies this (via the restriction map $r_{S \to M}$) to "scientists warn of health risk" for the chemical.
- The public interprets this (via the restriction map $r_{M \to P}$) as "toxic danger" from the chemical and responds with avoidance or political pressure.

Each transformation may appear locally consistent, in that each restriction map in this example is a plausible projection from a richer belief space to a simpler one. But globally, the system lacks a coherent global section: there's no consistent assignment of beliefs across all three nodes that respects every restriction map. This mismatch signals a structural inconsistency—not just misinformation as random noise, but as a *failure of compositionality* across patches, meaning that the local descriptions pieces cannot be joined (composed) into a shared coherent joint understanding of risk (global section). A sheaf-theoretic perspective reveals where this misalignment occurs and suggests how to fix it, for example, by redesigning $r_{S \to M}$ to better preserve uncertainty or context when communicating to the media.

Although information loss and distortion along a chain is mathematically trivial, this small example illustrates how loss of resolution or semantic compression along successive restriction maps can prevent reconstruction of a coherent global assignment that faithfully represents the source beliefs. For risk analysts,

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the point of casting S → M → P as restriction maps is not to rename an obvious phenomenon, but to clarify where and how distortions in risk messages and perceptions enter and how they might be removed. Each map is a potentially designable interface that transforms not only detail but the semantics of uncertainty (quantified → categorical → affective). Treating these as explicit mappings yields actionable levers: (1) we can specify invariants that must be preserved across maps (e.g., base rates, uncertainty bands, attribution conditions), (2) we can instrument the interfaces with audit checks (e.g., does "medium" at M still denote the same quantile band communicated by S?), and (3) we can consider how to redesign the maps (templates, metadata, constraints) so that downstream representations remain decision-relevant. In short, the sheaf framing turns a vague "loss of information through retelling" into a more testable and correctable interface problem. Despite its simplicity, the communication chain example is useful because it reframes "message drift" as a set of explicit, testable map designs: declaring which meanings must be preserved across S → M → P and checking them yields an approach for diagnosing and repairing risk distortion rather than merely describing it.

Recent updates to SARF (Kasperson et al. 2022) reflect the growing importance of digital and algorithmic communication platforms, which introduce feedback loops, content personalization, and accelerated message propagation. In these environments, traditional distinctions between information producer, amplifier, and recipient blur: lay users can become influencers; recommendation algorithms can reinforce pre-existing beliefs; and trustworthiness becomes contingent on perceived group identity or prior alignment. In these environments, misinformation can become not only more contagious, but also more self-reinforcing and harder to detect from within the system.

These abstract mechanisms of signal distortion and structural inconsistency are not just theoretical. For example, in one set of experiments, Moussaïd et al. (2015) examined how risk information—specifically, messages about the antibacterial agent triclosan—propagates through linear diffusion chains of human subjects. They found that while the factual content of messages degraded quickly across transmissions, the emotional signal of risk (i.e., whether a message conveyed danger or reassurance) was preserved and even amplified. Participants distorted content to match their own prior beliefs and then passed along messages that increasingly aligned with their subjective risk perceptions. As a result, neutral or even reassuring information could be transformed, through successive transmissions, into alarming narratives—especially in chains of like-minded individuals. The local mappings (individual-level belief transformations) may appear rational at each step, but the global result lacks coherence or fidelity to the original assessment. These findings offer empirical support for the idea that misinformation amplification is often a structurally emergent phenomenon, rather than a consequence of isolated cognitive errors.

3.3.2 | Example: Social Amplification of COVID-19 Vaccine Risk

To see how the preceding abstract ideas play out in real-world settings, consider the public discourse surrounding COVID-19 vaccines during early 2021. In this case, belief transmission involved not only institutional agents (scientific bodies, health agencies, traditional media), but also social media platforms, influencers, and algorithmic recommender systems, each of which acted as amplification stations with their own communicative logics.

Suppose we now expand the earlier network to include:

- S: a scientific advisory committee (e.g., CDC or WHO),
- M: mainstream media outlets,
- SM: social media platforms and user-influencers,
- P: the general public.

Each node is associated with a stalk:

- F(S): nuanced epidemiological models, vaccine trial data, statistical uncertainties.
- F(M): simplified expert interviews, headlines, and broadcast narratives.
- F(SM): viral posts, memes, emotionally charged anecdotes, engagement-optimized content.
- F(P): individual trust levels, perceived vaccine safety, and behavioral intentions.

The restriction maps now include:

- $r_{S \to M}$: Simplification of statistical nuance into expert soundbites or policy summaries.
- $R_{M \to SM}$: Algorithmic and social filtering, emotional reframing, and attention-grabbing distortions.
- $r_{SM \to P}$: Viral memes or personal stories that appear locally consistent but strip context, driving perceived danger, distrust, or political polarization.
- $R_{P \to SM}$ (feedback loop): Public reactions (likes, shares, comments) that reinforce the content visibility and affect what beliefs are fed back into the system.

For example, suppose that

- S assesses that vaccine-induced myocarditis is extremely rare, with rates of ~1 in 100,000 and typically mild cases.
- M reports this as "CDC Investigates Heart Inflammation Link to Vaccine in Teens."
- SM surfaces and spreads posts saying "COVID shot gave my cousin heart problems—be careful," amplified by algorithmic prioritization of controversy and emotional engagement.
- P interprets this cascade as "vaccines are risky," leading to hesitancy or refusal.

Each restriction map may appear locally justified: journalists condense scientific content; social users share emotionally salient experiences; algorithms optimize engagement. But there is no global section that represents all these communicated beliefs without contradiction. This view can help identify where and how corrective interventions should occur—for example, improving the $r_{S \to M}$ map by designing communication outputs explicitly

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

tailored for downstream virality-resistant clarity, or modifying $r_{M \to SM}$ by engaging platforms to carry context-preserving metadata with risk communications.

3.3.3 | Example: Hidden Semantic Ambiguity and Inconsistency in Risk Terms

As pointed out by Aven and Thekdi (2022), misinformation about risks can arise even in the absence of error or false reporting when terms are not clearly defined. A sheaf-theoretic perspective can help to clarify the sometimes subtle drift in how key risk terms are interpreted across agents.

For an example of how honest communication and pairwise agreement can nonetheless fail to yield a globally coherent interpretation due to semantic drift, consider a simple risk-communication network involving three actors: a scientific risk assessor (S), a media reporter (M), and a member of the general public (P). Each uses standard risk labels such as "Low," "Medium," and "High," but interprets them somewhat differently. Suppose that each actor interprets the label "Medium" as corresponding to a different range of risk values (where risk refers to the probability of a known adverse outcome): for S, "Medium" means a probability of harm between 0.30 and 0.50; for M, it corresponds to 0.40–0.60; and for P, the label denotes a risk between 0.55 and 0.70. These ranges reflect differences in institutional norms, communication styles, or risk thresholds. Each agent refers to a risk as "High" if it is above the range of "Medium" values and as "Low" if it is below the range of medium values.

Now imagine a communication process unfolding as follows. The scientific assessor S evaluates the probability of harm from a particular chemical exposure to be 0.45. Based on the "0.3–0.5" definition, S labels the risk as "Medium." This label is reported without alteration by M, who interprets "Medium" to mean 0.5 ± 0.1. P reads the article or hears the message from M and, likewise, accepts "Medium" as a meaningful label within P's own interpretive range—say, associating it with a value near 0.60. At no point is the message altered or disputed. Each step in the transmission chain is locally consistent: S's initial classification is accurate within its own model, M interprets and repeats the label in good faith, and P accepts it as credible. However, despite the absence of deception, error, distortion, or disagreement at any step, there exists a deeper inconsistency: no single numerical value lies in the intersection of all three "Medium" ranges. Visually, this can be imagined as three overlapping intervals whose pairwise overlaps are nonempty but whose triple intersection is empty:

S: [0.30–0.50] M: [0.40–0.60] P: [0.55–0.70]

Each adjacent pair shares common ground, yet no point satisfies all three—an everyday instance of a nontrivial H^1 that blocks formation of a global section. Thus, the system of beliefs, although pairwise compatible at each communication step, cannot be coherently "glued" into a single, shared global assignment that respects all the agents' constraints. In the language of sheaf theory, this is a failure of compositionality: the local sections (i.e., the individual beliefs and interpretations over overlapping pairs of agents) cannot be extended to a global section (i.e., a system-wide assignment of meaning) because of structural misalignment in the restriction maps. This illustrates a *semantic* incoherence: misalignment of meaning even when content and trust remain intact.

Although this example is simple enough that no advanced mathematics is needed to understand why coherence fails, it is instructive to see how the same situation appears formally within the sheaf framework. Each agent's interpretation of *Medium* is its local data (its stalk):

$$F(S) = [0.30,\ 0.50]\,, F(M) = [0.40, 0.60]\,, F(P) = [0.55,\ 0.70]\,.$$

For any triple

$$x = (x_S, x_M, x_P) \in F(S) \times F(M) \times F(P)\,,$$

representing the numerical values (quantitative beliefs about risk) that S, M, and P associate with the qualitative label *Medium* for a hazard, local consistency on an overlap (e.g., edges S–M or M–P) requires that the two agents agree on the value of *Medium* within their shared semantic context (corresponding to the edge between them). If all agents are truthful, as assumed in this example (and in the opinion-dynamics models previously reviewed), then the restriction maps are simply identity maps (i.e., each agent discloses a value that is its true belief), so pairwise consistency of meanings within the discourse spaces for the edges is the condition that $x_S = x_M$ on edge S–M and that $x_M = x_P$ on edge M–P.

$$x_S = x_M \text{ on edge SM},\ x_M = x_P \text{ on edge MP}.$$

But for such a globally consistent assignment to exist, the value θ must lie in all three stalks simultaneously:

$$\theta \in F(S) \cap F(M) \cap F(P)\,.$$

In this example, however,

$$F(S) \cap F(M) \cap F(P) = \emptyset,$$

so no global section exists. That is, no numerical value satisfies all three agents' interpretations of "Medium."

This same conclusion emerges from the coboundary operators introduced earlier. The sheaf on this three-node chain has three nodes (S, M, P) and two edges (SM, MP), so the node–edge coboundary matrix is

$$D^0 = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

A triple (a "0-cochain") $x = (x_S,\ \ x_M,\ \ x_P)$ is locally consistent when its edge-difference vector $\omega = D^0\, x$ lies in the kernel of D^1. Since the S–M–P network contains no closed loop (no triangle), there are no higher order overlaps of edges, so $D^1 = 0$. Thus, local consistency requires

$$D^0 x = 0 \Leftrightarrow\ x_S = x_M = x_P.$$

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Therefore, a global section must assign the *same* value θ to S, M, and P. Since no such θ lies in all three stalks, we again conclude that the set of global sections is empty. Equivalently, the space of locally consistent edge assignments contains elements that are not the restrictions of any global assignment. In the linear cohomology framework of Section 3.2, this means the first cohomology group is nontrivial:

$$H^1(F) \neq 0.$$

This calculation shows that the obstruction is structural, not accidental: the geometry of the three intervals forces local pairwise compatibility without permitting global coherence. Sheaf theory makes this failure explicit and measurable, identifying precisely where semantic alignment breaks down even when all communication is honest, accurate, and mutually trusted.

In this example, imprecise semantics (here, interval-valued meanings of the term *Medium*) integrate naturally with the quantitative sheaf machinery. The intervals supply the local data (stalks), the restriction maps (here, identity maps for truthful agents) encode how meanings must align on overlaps for pairwise consistency, and the cohomology calculation then provides a diagnostic showing that these locally reasonable interpretations cannot be composed into a coherent global meaning. Although the sheaf formalism is not needed for such a simple example, applying it shows how the qualitative content—the agents' different implicit meanings of *Medium*—is represented by the stalk intervals. The restriction maps formalize how these meanings must align where agents overlap. The coboundary operators D^0 and D^1 then convert those qualitative constraints into a concrete linear–algebraic test of compositionality. The resulting nontrivial H^1 encodes the precise structural reason why truthful, accurate, locally compatible messages nevertheless fail to yield a coherent global assignment of meaning. In this way, the sheaf-theoretic machinery sharpens what intuition already suggests, turning qualitative concerns about "misalignment of meanings" or "semantic drift" into explicit structural tests for when local agreements between agents fail to compose into a globally coherent belief system.

This example shows that, when there are unstated assumptions and definitions, even accurate and honest risk communication can generate semantic inconsistency that is invisible to traditional probabilistic or behavioral models (Aven and Thekdi 2022). It also suggests how a sheaf-theoretic framework can diagnose failures of global understanding even when local trust, accuracy, and honesty are preserved. This mechanism for the spread of misinformation does not depend on agents adjusting their beliefs to each other, as in the opinion-dynamics models discussed previously, but on their failure to adjust beliefs to account for differences in meaning when the same terms are used by different agents. This type of failure may be especially important in the spread of beliefs about whether exposures cause human health harms. Assertions such as "Y deaths per year are attributable to exposure X" that have a specific meaning to epidemiologists are widely misreported and misconstrued as meaning "Y deaths per year could be prevented by removing exposure X," although specialists recognize that neither implies the other (Cox 2019).

More generally, in misinformation-rich environments, global sections may fail to exist not because of filtering, distortion, or incompatible belief transformations and semantic interpretation of labels across agents. For instance:

- A public health agency uses an ordinal scale for risk (low/medium/high), while local governments use color codes (green/yellow/red), and community influencers summarize beliefs as binary decisions (safe/unsafe). Even if each mapping is internally valid, the system may not admit a consistent global representation. This type of failure is illustrated in the above examples.
- An individual's private belief (e.g., that a vaccine is safe) is softened when communicated to a skeptical peer group. The softened message is then amplified and reframed as uncertainty by others. The original belief and its public echoes can no longer be reconciled. This type of failure is discussed more carefully next.

Both of these are examples of restriction maps failing to preserve key structure, leading to global misalignment—even if no actor is acting maliciously or irrationally. Such phenomena are difficult to capture in traditional network models but are a natural focus for sheaf-theoretic analysis.

3.4 | Opinion Dynamics With Misrepresented Opinions: Sheaf Laplacians

Traditional models of opinion dynamics such as the DeGroot, FJ, HK, and DW models use a single number to represent an individual's belief at any moment. They model how these beliefs adjust to each other over time in response to the differences between them (Figures 1–4). A useful generalization for modeling misinformation and disinformation is to allow opinions to be represented by vectors rather than by single numbers. This is important for modeling the distinction between true opinions and expressed opinions, for example. This section describes how sheaf theory accommodates this generalization and also how it models the adjustment of these richer beliefs over time using sheaf Laplacians.

The sheaf Laplacian framework (Hansen and Ghrist 2021) provides a more expressive and structurally explicit approach to modeling opinion dynamics in complex networks. It generalizes classical models by allowing each agent to hold multidimensional or structured beliefs—represented not by a single number, but by a vector or more general data type—and by explicitly modeling how these beliefs are expressed to others in the network. Each agent's private belief lives in a local vector space (its *stalk*), and the belief the agent expresses in a particular interaction is derived from this private belief via a *restriction map*, which may differ from one communication context to another. The basis of the agent's opinion space is typically a set of topics (such as those used in questionnaires in the cultural theory of risk, e.g., about perceived risks from climate change, GMOs, nuclear power, etc.) and the vector representing a private opinion shows how an agent feels about each topic, from very negative through neutral to very positive.

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

This added flexibility is especially important when modeling misinformation, disinformation, and semantic misalignment. In traditional models such as DeGroot, beliefs are assumed to be fully and faithfully communicated: agents observe neighbors' opinions and average them directly. In contrast, the sheaf-theoretic model allows for selective expression, framing, simplification, distortion, or even intentional misrepresentation. These effects are captured in the structure of the restriction maps. For example, an agent may convey a softened version of its beliefs to a skeptical peer group while expressing stronger convictions to allies. Such differences in communication strategy are not easily accommodated in scalar-valued models.

The dynamics induced by the sheaf Laplacian are defined analogously to classical diffusion processes: beliefs evolve over continuous time to reduce local disagreement in expressed opinions. However, rather than driving all opinions to a common value, these dynamics seek a *harmonic state*—a configuration in which each agent's expressed belief is consistent with those of their neighbors, as determined by the network's restriction maps. Importantly, agents' private beliefs may remain different, and in fact may not even be directly comparable across the network. What matters is that expressions of belief, once mapped into shared discourse contexts, align in a way that respects the local structure of the system. In other words, the objective function to be minimized is assumed to be differences between *expressed* opinions rather than differences between *true* (private) opinions.

To define how these richer beliefs evolve over time, the discourse sheaf-theoretic framework defines the sheaf Laplacian as an operator (acting on the current state of private beliefs) that determines how disagreements among agents' current expressed beliefs drive changes in their private beliefs. The idea is simple: if neighboring agents express beliefs that do not align—after accounting for how each expresses beliefs in that particular context—then the sheaf Laplacian models how those misalignments are reduced over time. Formally, the sheaf Laplacian is defined as $L = (D^0)^t D^0$, where D^0 is the first *coboundary matrix*, introduced earlier. Each row of D^0 corresponds to a point of communication or overlap between two agents—such as a shared conversation, a reporting relationship, or a common interpretive framework—and each column corresponds to an agent's own belief domain. D^0 captures differences in how agents express their personal (private) beliefs across overlapping discourse contexts after they are appropriately translated by restriction maps. These maps describe how each agent's private belief is expressed in a specific shared context, such as simplifying technical details for a general audience or using emotionally salient framing in social media. Beliefs expressed by neighboring (communicating) agents should align when appropriately translated into their shared discourse context.

When two agents disagree in their expressed beliefs—whether because of different private beliefs or because of differences in how those beliefs are communicated—this disagreement appears as a nonzero value in the residual vector $\omega = D^0 x$, where x is the vector of current private beliefs and $D^0 x$ represents the differences in how those beliefs are expressed in shared discourse contexts, as translated via the restriction maps. The sheaf Laplacian L then determines how strongly each private belief is "pulled" toward better alignment with its neighbors. This leads to a continuous-time dynamic equation: $dx/dt = -\alpha\, Lx$, where α is a positive rate parameter. Under this rule, each agent's private belief gradually shifts in the direction that reduces disagreement in expressed messages across the network. The sheaf Laplacian framework explicitly incorporates the possibility that messages are framed, simplified, or distorted via context-dependent translation rules. The process continues until the system reaches a state of minimal tension—if such a state exists—where all expressed beliefs are mutually compatible. This equilibrium, known as the *harmonic state*, generalizes the idea of consensus: rather than requiring all agents to adopt identical beliefs, it requires only that their expressed beliefs, once translated into shared discourse spaces, are mutually consistent, even if private beliefs differ. In a harmonic state, the sheaf Laplacian is 0—that is, $Lx = 0$—and no further adjustment takes place.

When no such globally coherent configuration exists, the sheaf Laplacian framework can detect and diagnose these failures using the first cohomology group H^1: $\dim(H^1) > 0$ indicates the presence of differences in expressed beliefs that cannot be resolved, no matter how agents adjust their private opinions. Shared understanding may break down even in the absence of noise, bias, or irrationality simply because of overlapping but misaligned interpretive frameworks, semantic ambiguities, or inconsistent translation rules across institutions or communities. Classical models of opinion dynamics, such as the DeGroot model and its variants, are not equipped to model such failures, as they assume a single shared belief space.

3.4.1 | Example: Pareto-Inefficient Harmonic Agreement in an Impoverished Discourse Space

The sheaf Laplacian framework tracks how agents' private beliefs, expressed through context-dependent restriction maps, evolve toward local consistency. When all expressions of belief are mutually compatible in their shared discourse contexts, the system reaches a harmonic state—that is, one in which the sheaf Laplacian vanishes ($Lx = 0$) and no further adjustment occurs. However, this harmonic state need not be collectively desirable. The following example shows that when the discourse space in which beliefs are expressed and compared is too limited to preserve the meaning or source of underlying beliefs, the result may be a misguided consensus leading to outcomes that no one wants.

Suppose a panel of three analysts—A, B, and C—is tasked with evaluating a proposed environmental regulation. The policy has consequences in three dimensions of concern: health (H), economic impact (E), and environmental quality (B). Each analyst holds private beliefs about these dimensions and communicates support or opposition to the policy through an overall support score. This scalar constitutes the shared discourse space—a one-dimensional space into which multidimensional private beliefs are projected. Initially, all agents share the same prior belief: that the regulation will have beneficial effects on all three dimensions, with expected values of +1.0. Each agent then receives accurate private information ("expertise") about a different outcome dimension, revealing that the regulation is harmful in that domain. Specifically, agent A observes that the policy will worsen

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

health outcomes ($H = -1.0$); agent B learns that it will harm the economy ($E = -1.0$); and agent C discovers that it will degrade environmental quality ($B = -1.0$). Each agent assumes that the other two dimensions remain as previously believed (i.e., at +1.0), since no information to the contrary is available. This results in the following private belief vectors, each defined over a three-dimensional local stalk:

- A: [−1.0, +1.0, +1.0]
- B: [+1.0, −1.0, +1.0]
- C: [+1.0, +1.0, −1.0]

Each agent then expresses its belief in the shared discourse space via a restriction map—a projection from the agent's three-dimensional private belief space to a scalar support score, obtained by uniformly averaging the three components. Each reports a support score of $(+1 + 1 - 1)/3 = +0.33$.

These expressed beliefs are locally consistent: they are identical. All three agents express the same scalar support value, despite holding distinct and mutually incompatible private beliefs. This perfect alignment in expression conceals irreconcilable differences in the underlying justifications for support, which are invisible in the compressed discourse space. No agent disagrees and none finds reason in the expressed opinions of others to modify its beliefs, so the situation is harmonic: the expressed beliefs, as translated via the restriction maps, are mutually compatible in the one-dimensional discourse space, leading to a vanishing sheaf Laplacian, $Lx = 0$. Yet this alignment is deeply misleading. The agents have failed to communicate which component of the outcome vector each has evaluated. The restriction maps erased this structure, leaving each agent free to assume that the mild support expressed by others reflects shared, well-informed optimism across all dimensions. As a result, all agents infer that the regulation must be broadly beneficial, and the group collectively endorses its adoption.

Once implemented, however, the policy's true consequences are revealed: it is harmful in every dimension. Each agent is dismayed to discover that the domains they presumed were favorable were, in fact, those others had observed to be harmful. If the private information had been shared in full—or even just labeled by dimension—the group would have reached a very different conclusion. Instead, the impoverished discourse space and structure-erasing restriction maps created a perverse consensus of expressed beliefs, masking irreconcilable contradictions in the underlying beliefs. From a sheaf-theoretic standpoint, the system's belief sheaf admits a harmonic section in the reduced discourse space, but that section is ex post Pareto-dominated: there exists no agent who prefers the actual outcome to the one that would have resulted from truthful belief sharing. The problem is not a lack of convergence or coherence, but a failure of expressivity: the structure of the discourse prevented critical distinctions from being preserved.

This example shows that harmonicity does not imply Pareto-efficiency. When critical distinctions—such as which belief component was informed by direct observation—are lost, even perfect agreement in expression may lead to collectively regrettable outcomes. One remedy is to enrich the discourse space—for example, by explicitly tagging expressed support with which dimensions it reflects, or by allowing agents to communicate partial belief vectors rather than scalar summaries. Doing so expands the shared structure over which beliefs can be integrated and increases the likelihood that global coherence reflects true agreement, not merely projected alignment. In this way, sheaf-theoretic analysis provides not only diagnostics for when communication fails, but also design principles for more expressive and robust systems of shared belief and decision support.

3.4.2 | Case Study: Quantitative Effects of Deceptive Expression in Sheaf-Laplacian Opinion Dynamics

The preceding example illustrated how honest but structurally constrained communication can yield misleading consensus even when all participants act in good faith. A complementary challenge arises when some agents deliberately misrepresent their private beliefs—spreading *disinformation* rather than mere *misinformation*. This case study illustrates how systematic distortion in expressed beliefs alters influence and equilibrium outcomes through the structure of communication, rather than through network topology alone.

The sheaf-theoretic framework accommodates the case of deliberately distorted communications naturally: deceptive communication corresponds to nonidentity restriction maps that systematically distort how private beliefs are expressed in shared discourse spaces. In this view, each agent i holds a private belief vector x_i in its local stalk and communicates an expressed belief $r_i(x_i)$ to its neighbors via a restriction map r_i. Truthful agents use identity maps ($r_i(x_i) = x_i$), while deceptive agents use biased or strategically crafted maps that reshape their messages—for instance, by exaggerating risks or downplaying uncertainty. When such messages propagate, the resulting residual vector $\omega = D^0\ x$ contains structured, rather than random, inconsistencies. These systematic distortions generate nontrivial first cohomology ($H^1 \neq 0$), revealing global inconsistency that cannot be removed by local belief updates. Consequently, the sheaf Laplacian converges not to a truthful harmonic equilibrium, but to one aligned with the biased subspace defined by deceptive restriction maps.

This dynamic parallels the quantitative findings of Aktas et al. (2024), who examined how *intentional deception* alters influence in empirical social networks. Even a small fraction of "liar" nodes—agents whose expressed messages systematically diverge from their private beliefs—can exert disproportionate sway when occupying structurally central positions. Because deceptive messages often appear confidently precise or emotionally salient, they attract alignment responses and shift the system's equilibrium toward the deceptive subspace. In sheaf-theoretic terms, the liars' biased restriction maps introduce persistent cohomological obstructions: locally plausible but globally irreconcilable belief configurations. These obstructions redirect diffusion flows in the sheaf Laplacian toward distorted yet internally consistent equilibria.

To illustrate more concretely how sheaf theory can clarify and quantify the structural effects of misinformation in social networks, Aktas et al. (2024) used realistic network data to

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Empirical social networks spanning diverse mechanisms of interaction used to test effects of deceptive communication.

Dataset	Domain/type	Typical structure cues	Example source
Jazz	Collaboration among jazz musicians and bands; real coproduction contexts with strong community/clique structure	Dense, high clustering, overlapping communities	Gleiser and Danon (2003)
Netscience	Coauthorship among network-science researchers; professional collaboration and community boundaries	Sparse, modular, heavy-tailed degree distribution	Newman (2006)
Les Misérables	Narrative coappearance of characters; interpretable social structure	Medium density; clear mesoscale communities	Knuth (1993)
David Copperfield ("Copper")	Word-adjacency (adjective–noun) network; extends analysis beyond interpersonal graphs	Small-world features; linguistic constraints	Newman (2006)
Highschool	Self-reported friendships among boys in an Illinois high school (1957–1958)	Directed/weighted variants; reciprocity and temporal persistence	Coleman (1964)
Oz (ANU residence hall)	Friendship ratings among 217 university residents	Relatively dense; high neighborhood overlap; weighted ties	Brewer (2000)
Congress	Legislative cosponsorship network in the US Congress; institutional coalitions and party structure	Community structure along partisan lines	Fowler (2006)
Innovation	Advice/communication among physicians in Illinois towns during tetracycline adoption; canonical diffusion study	Medium density; directional and weighted ties	Coleman et al. (1957)
Train	Reconstructed associations among suspects in the 2004 Madrid bombing; clandestine social network	Medium density; multiplex tie strengths	Rodríguez (2004)

analyze how deceptive communication, modeled as biased expression functions (restriction maps) on network edges, alters influence and consensus formation within the framework of sheaf-Laplacian opinion dynamics. The analysis draws on nine well-established empirical social networks spanning diverse mechanisms of interaction: collaboration, friendship, narrative coappearance, institutional coalition, diffusion of innovation, and clandestine association. These datasets, used as benchmarks in network science, span dense and sparse regimes, directed and undirected edges, weighted and unweighted ties, and networks with both strong and weak community structure, making them useful for exploring how deception interacts with network topology.

Table 3 summarizes the nine benchmark network datasets. The cited references in the right column and Aktas et al. (2024) provide more details. The networks are intended to be diverse enough to demonstrate that the effects of deceptive expression are important in networks ranging from the tightly knit Jazz collaboration network to the sparse, modular Netscience graph, that is, for a variety of social contexts where misinformation or strategic distortion can arise.

Within each network, nodes represent agents holding private scalar opinions x_i. Honest agents communicate these opinions faithfully ($y_i = x_i$), whereas deceptive agents transmit *biased expressions*:

$$y_i = \alpha_i\, x_i + (1 - \alpha_i)\, b_i,$$

where $\alpha_i \in [0, 1]$ is the *honesty parameter* ($\alpha_i = 1$ = truthful; $\alpha_i = 0$= fully deceptive) and b_i is a *strategic bias term* indicating the direction and magnitude of intentional distortion. The bias b_i can represent, for example, exaggeration of risk to provoke alarm, understatement of uncertainty to project confidence, or selective framing to align messages with anticipated audience preferences. In general, b_i captures any systematic shift between an agent's *private belief* and its *publicly expressed message* designed to influence others' perceptions or consensus formation. In the sheaf formulation, these expression functions define the *restriction maps* that link private beliefs to communicated beliefs. Varying these maps while holding network topology constant isolates how *misexpression alone*, not connectivity or degree, affects diffusion of influence.

In these models, each agent repeatedly adjusts its belief by comparing it with the beliefs expressed by its neighbors and then moving partway toward local agreement in *expressed* beliefs. Mathematically, this process can be written as a system of differential equations, $\dot{x} = -L_{\mathcal{F}} x$, where the sheaf Laplacian matrix operator $L_{\mathcal{F}}$ is a linear (block matrix) operator built from the restriction maps that governs how information differences

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

are smoothed (averaged) across the network. In ordinary network models (using the simpler *graph Laplacian*), this equation just means that connected agents gradually average their opinions until disagreement disappears. The sheaf version generalizes this idea by allowing each agent to express information in its own units or terms. When some agents distort their messages, the adjustment process converges to a biased equilibrium that reflects the structure of the deceptive communications rather than the underlying truth.

The influence of individual agents within this framework can be summarized by the following two complementary *centrality measures* that together describe how information flow and disagreement resolution are distributed through the network.

- *Sheaf-Laplacian centrality* generalizes conventional graph-based centrality to incorporate how agents express and compare information through the restriction maps. It quantifies the extent to which perturbing an agent's expressed message alters the network's overall reduction of expressed disagreement. Higher values indicate that the agent's communications exert stronger leverage over the trajectory of diffusion and the position of the eventual consensus in expressed beliefs.
- *Sheaf-Diffusion-Flow-Field (DFF) centrality* provides a complementary flow-based perspective, reflecting how much of the adjustment current (i.e., the flow of incremental opinion changes induced by the update rule $\dot{x} = -L_{\mathcal{F}}x$ as they propagate along edges via the restriction maps) generated by the sheaf-Laplacian dynamics passes through each agent. Nodes that lie on major information pathways or communication bottlenecks exhibit low DFF values, signifying high control over the routing of influence.

These measures are computed from the same network topology but differentially weight the restriction maps; they therefore isolate the effects of *how* agents communicate from the underlying pattern of *who* communicates with whom. Because the graph structure is held fixed in the analyses of Aktas et al. (2024) and Aktas et al. (2024), changes in these metrics arise solely from modifications in the expression functions (e.g., lying or distorting beliefs when expressing them to others).

Absolute magnitudes are not directly comparable across networks of differing size or density, so results are interpreted in terms of within-network contrasts: ratios or percentile shifts between deceptive and honest agents. As the honesty parameter α decreases, deceptive nodes generally exhibit higher Sheaf-Laplacian centrality, while DFF responses depend more on network density, with dense graphs partially attenuating liar advantages. Elevated liar centralities often coincide with nontrivial first cohomology ($H^1 \neq 0$), indicating persistent inconsistency in the global belief configuration, although the relation is not one-to-one: centrality captures influence, whereas cohomology diagnoses consistency.

Both quantities are computable from expressed messages and known restriction maps, without access to private beliefs, making them operational for observed communication data. Their interpretation, however, is structural rather than normative: high centrality denotes influence, not correctness. In practical terms, modifying restriction maps—through reweighted trust matrices, message-verification metadata, or redundant information channels—can lower deceptive agents' Laplacian centrality or raise their DFF values, reducing their ability to steer collective beliefs.

Across the nine datasets, deceptive agents exhibit striking and consistent influence advantages. Representative results include the following influence statistics, measured by Laplacian centrality, for honest versus strategic ("liar") communicators in three networks:

Network	Honest	Liar (prosocial)	Liar (antisocial)	Ratio (liar:honest)
Train	0.380	1.138	1.131	≈3×
Highschool	0.279	1.012	1.005	≈3.6×
Netscience	0.016	0.170	0.170	≈10.6×

Averaged over all nine networks, total Laplacian-centrality mass equals 0.142 for honest agents versus 0.565 for liars—roughly a fourfold advantage for liars. Under the DFF metric, results depend on network density: liars dominate in seven of nine networks, but in the densest graphs (Jazz ≈ 27.7, Oz ≈ 16.9 average degree) honest agents perform slightly better. Varying the honesty parameter α from 0 to 1 narrows but never eliminates the liar advantage under Laplacian centrality.

The main finding in these networks is that, holding everything else fixed, altering only the communication maps can increase an agent group's measured influence by up to an order of magnitude. Such findings have direct relevance for risk communication, public health messaging, and social media governance, where distortions of perceived credibility or selective amplification can alter system-level behavior even without explicit falsehoods.

4 | Intervening to Reduce Harm From Misinformation: Diagnosis, Design, and Control

Much of the prior literature and policy guidance on combatting misinformation and disinformation focuses on correcting individual claims and adjusting individual beliefs. The sheaf-theoretic perspective prompts an underlying question as well: under what conditions can a distributed system of beliefs be coherently integrated at all? Sheaf theory addresses this question by modeling not only beliefs, but also how they are expressed, shared, and communicated across a network of agents, each with potentially different perspectives, vocabularies, and informational constraints.

This section examines how the sheaf-theoretic perspective informs the design of interventions—both corrective and preventive—that reduce the harms of misinformation. We emphasize three recurring leverage points: changes to the *structure* of communication, adjustments to the *semantics* of belief expression, and *targeted use of control* (e.g., injection of corrective messages) and observability (monitoring the state of

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

misinformation) to stabilize or realign belief dynamics. While not all interventions fit neatly into a single category, this framework helps organize the underlying mechanisms by which coherence can be disrupted or restored in complex belief systems.

4.1 | From Inconsistency Detection to Actionable Diagnosis

When global sections fail to exist (i.e., $Lx \neq 0$ for all x), the cohomological structure points to which overlaps (e.g., shared discourses, institutions, or belief junctions) are responsible. In such cases, interventions might focus on bridging disconnected subnetworks, modifying restriction maps to reduce semantic drift, or inserting corrective information at structurally central nodes. This diagnostic capability enables the transition from passive identification of misinformation to active fault localization and repair—analogous to fault-tolerant design in engineering systems or model-based diagnosis in artificial intelligence.

Traditional misinformation interventions, such as fact-checking, content moderation, and algorithmic suppression, aim to correct false claims or steer behavior toward desired outcomes. Yet empirical studies show that these approaches often fall short, especially when trust is fragmented or beliefs are tied to social identity. Backfire effects, selective exposure, and emotional resistance all contribute to the persistence of misinformation, even when credible corrections are available. From a sheaf-theoretic standpoint, these difficulties arise when the belief network's structure itself impedes information integration. Even accurate content may fail to alter the global belief landscape if it cannot be coherently composed with existing local beliefs. In such cases, the failure is not cognitive, but architectural: the system lacks the expressive or semantic structure needed to absorb corrections. This insight suggests that effective interventions must target not just messages, but the conditions under which messages can be meaningfully compared and integrated.

4.2 | Controllability and Stabilization of Belief Dynamics

Hansen and Ghrist (2021) develop a control-theoretic extension of sheaf-based opinion dynamics that formalizes how belief systems can be steered toward coherence. By modeling belief change using a continuous-time differential equation ($dx/dt = Lx + Bu$), where B represents an input matrix and u the controller's influence on a subset of agents, they demonstrate that the entire network can be guided to a desired belief configuration if the relative cohomology $H^0(G_U, F)$ vanishes.

This result provides a powerful and rigorous foundation for strategic intervention. It implies that by influencing only a small, well-positioned subset of agents, one can realign the belief dynamics of the entire system—without requiring complete control or universal agreement. A dual result for observability shows that if $H^0(G_Y, F) = 0$ for an observation subset Y, then the global state can be inferred from local measurements, allowing for real-time monitoring of emerging incoherence.

This observer–controller paradigm offers a roadmap for misinformation countermeasures that are minimally invasive, structurally informed, and scalable. Rather than trying to correct everyone, it may suffice to influence or monitor key actors positioned at critical junctions in the communication graph—those whose restriction maps anchor or distort downstream belief propagation.

A complementary but distinct approach is developed by Jamalzadeh et al. (2022), who integrate an epidemiological SIR model of disinformation spread with a network optimization model of infrastructure flow. Their goal is to minimize system-wide commodity shortages caused by behavior changes induced through misinformation—for instance, surges in electricity demand following a fake price discount message. By modeling the effect of disinformation on user consumption and integrating it with critical infrastructure constraints, they formulate a mixed-integer optimization problem that selects both information countermeasures and system-level adaptations. Although their approach does not use sheaf-theoretic language, it embodies a structurally informed control strategy: diagnosing where false beliefs disrupt physical systems, and countering them by targeting corrective messages at structurally influential nodes. This optimization paradigm parallels the sheaf-theoretic observer–controller framework by emphasizing structurally minimal, strategically positioned interventions that realign system behavior and performance (Jamalzadeh et al. 2022).

4.3 | The Role of Discourse Design and Semantic Interoperability

Another implication of the sheaf model is that coherence cannot be restored unless the discourse space is rich enough to represent the distinctions that matter. When agents compress their multidimensional private beliefs into a shared one-dimensional support score—such as "approve" or "oppose"—they may converge on a harmonic state ($Lx = 0$) that is deeply misleading (Hansen and Ghrist 2021). As illustrated in Section 3.4, such compression can lead all agents to endorse a policy that, once implemented, none of them supports. The problem in this case is not divergence, but insufficient expressivity. The sheaf Laplacian guarantees only that expressed beliefs are locally consistent, not that they reflect globally desirable outcomes. To address this, interventions may need to focus on expanding the shared discourse space or redesigning restriction maps to preserve more semantic content. Examples include tagging beliefs with evidentiary provenance, distinguishing between epistemic and affective components, or aligning terminologies across institutions and platforms.

4.4 | Examples of Structure-Aware Interventions to Combat Misinformation

This section presents several empirical and modeling examples illustrating the key sheaf-theoretic insight that effective interventions against misinformation depend not only on the content of corrective messages, but perhaps even more on the structure of belief networks and the placement of trusted communicators. These diverse cases show that restoring global coherence in belief systems requires attention to semantic alignment, trust pathways,

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

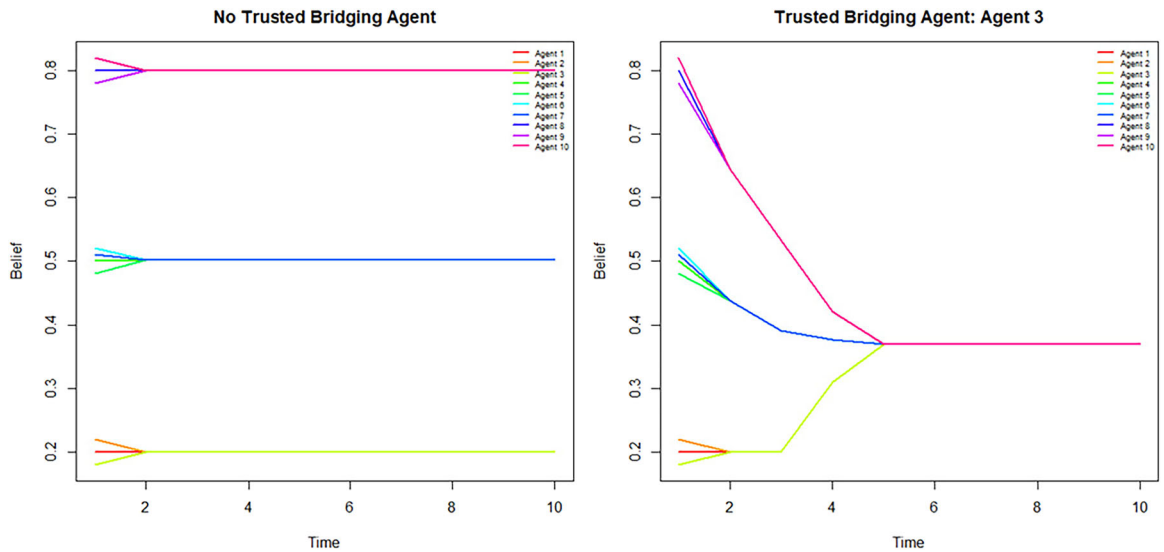

FIGURE 5 | Restoring belief coherence through a trusted bridging agent in a polarized network.

and the topology of communication flows. These studies may be interpreted as providing empirical and simulation-based support for structural diagnostics and intervention strategies inspired by the mathematics of gluing and cohomology.

Several recent empirical studies and simulation models support this structural view. In Thailand, a detailed doctoral study of food-related disinformation on social media found that science social influencers played a central role in debunking viral food rumors and restoring trust in scientific information (Pitigraisorn 2020). These "super-correctors" combined reputational credibility, subject-matter expertise, and wide social media reach to counter misinformation within belief clusters. Although not framed in mathematical terms, the study's findings suggest a sheaf-theoretic interpretation in which such actors enabled gluing by bridging otherwise disconnected discourse communities and promoting semantic alignment. Their role illustrates how structurally embedded trust agents can restore coherence to fragmented public belief systems.

4.4.1 | Example: Bridging Polarization in Risk Beliefs Through a Trusted Communicator ("Super-Corrector")

Reflecting the "super-corrector" role, Figure 5 illustrates two simulations of belief dynamics in a simplified model of how individuals update their risk perceptions through communication. Each of the 10 agents starts with an initial belief about a risk, such as the safety of a food product or the reliability of a public health recommendation. These beliefs fall into three distinct clusters—low, medium, and high concern—reflecting a polarized population. The agents update their beliefs over time using a bounded confidence (HK-type) model, meaning they only take into account the views of others whose opinions are sufficiently close to their own (within an "epsilon" range of 0.2).

In the left panel, no special communicator is introduced. Each agent updates its belief only based on the beliefs if its neighbors and only if their beliefs are within 0.2 of the agent's belief. In this case, the agents converge to three separate clusters, with no system-wide agreement. Belief clusters form and persist, and no consensus is reached. As in Figure 4, this mirrors many real-world situations in which polarized communities remain fragmented despite repeated exchanges of information.

By contrast, in the right panel, a single "trusted bridging agent" or "super-corrector" is introduced. This agent is treated as edible by all others, even if their beliefs are initially far apart. The other agents update their beliefs by including the trusted agent's view, even when they would normally disregard it due to the confidence bound. This reflects situations where a widely trusted source (e.g., a national health authority, a scientific communicator, or a respected news figure like Walter Cronkite) has asymmetric influence. In the right panel, trust in this single bridging agent enables all agents to converge to a shared belief, even though their initial fragmented beliefs would otherwise have become three separate echo chambers. This occurs without any change in the agents' own bounded confidence thresholds: just the asymmetric trust in one communicator is enough to restore coherence.

This example shows that structural interventions—such as placing a trusted communicator in a central role—can eliminate polarization that would otherwise persist indefinitely. From a sheaf-theoretic standpoint, the bridging agent enables gluing of locally consistent but globally disconnected belief clusters, restoring the possibility of a coherent global section. For risk analysts, this highlights the importance of not just the content of risk communication, but the network structure of trust and information flow. In fragmented systems, coherence must be designed, not assumed—and trusted bridging agents may be a key part of that design.

Important as the structure of communications networks and trust is, as illustrated in Figure 5, the *content* of messages—including their emotional valence—is also crucial in driving the virality and spread of misinformation. For example, during the 2025 California wildfires, a large-scale analysis of high-engagement YouTube comments revealed that emotionally salient

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

misinformation—including conspiracy theories about water mismanagement, diversity initiatives, and directed energy weapons—gained meaningful traction, despite constituting less than 1% of total discourse (Erokhin 2025). Comments referencing these themes displayed disproportionately high levels of anger and sadness, often rooted in distrust of authorities and perceptions of systemic failure. These findings support the view that locally coherent but semantically misaligned clusters of belief can persist and polarize public discourse. From this perspective, coherence was obstructed not merely by factual inaccuracy, but by the absence of a shared dispassionate semantic framework capable of integrating divergent interpretations into a unified factual narrative about risk and uncertainty.

A related simulation study by Tambuscio and Ruffo (2019) explored strategies for limiting the spread of urban legends in segregated social networks by introducing "eternal fact-checkers"—agents that persistently debunk misinformation and never revert to a susceptible state. Their model showed that strategically placing such agents—especially in high-degree nodes (hubs) or along the frontier between segregated communities—could significantly reduce the equilibrium level of belief in false information, even under pessimistic assumptions where individuals never verify information themselves. These findings support the concept of structure-based interventions: the success of debunking depended less on changing individual minds and more on where corrective signals were placed within the communication graph. From a sheaf-theoretic perspective, such agents can be reinterpreted as *structural anchors* that modify belief propagation pathways in ways that restore the possibility of global coherence.

The importance of location of (dis)information sources has also been studied by Jamalzadeh et al. (2022) in simulations of the spread of price-related disinformation affecting electricity use in Los Angeles County. Their hybrid model identifies locations where behavioral responses to misinformation would likely cause shortages and overloads in the power grid. By optimizing both the flow of electricity and the placement of counterinformation, they demonstrate that modest interventions—targeted at key nodes—can significantly reduce infrastructure risk.

An insight from these models and case studies is that effective risk communication requires more than persuasive messaging or behavioral nudges. It also requires communication structures that support recoverability: the ability of a belief system to integrate new, potentially corrective information without fragmenting or polarizing. Sheaf-theoretic models provide both the diagnostics and the control strategies to evaluate and enhance this capacity. This may involve redesigning reporting relationships, aligning semantic vocabularies, fostering cross-cutting affiliations, or selecting bridge agents who can navigate multiple discourse communities. It also involves understanding where structural obstructions to gluing—such as missing trust relationships, semantic drift, or asymmetric restriction maps—are most acute. This structural approach does not replace content-level fact-checking, but potentially increases its effectiveness. By modeling the architecture of belief expression and transmission, it becomes possible to diagnose not only what is believed and by whom, but why certain systems resist correction even when all individual agents are sincere and attentive. This reframing, we argue, is essential for risk analysis in an era where the harms of misinformation are increasingly systemic rather than sporadic.

5 | Discussion: Current Conclusions and Future Directions

Misinformation and disinformation pose persistent challenges to the coherence, reliability, and responsiveness of modern risk-communication systems. Although it has long been widely recognized that risk communication involves multiple groups with partially overlapping and sometimes inconsistent information, the sheaf framework provides explicit mathematical representations of each group's local knowledge (stalks), the relationships between groups (restriction maps), and the conditions under which locally coherent communications can be composed into a globally coherent belief state. Cohomology then supplies computable diagnostics for when such composition is impossible. In this way, the approach does not merely replace or restate qualitative insights in more technical terms, but sharpens them, turning intuitive notions of overlap, drift, or misalignment into precise structural tests that explain where and why global coherence fails.

Sheaf theory offers several potentially useful perspectives for risk analysts. First, it underscores that global incoherence in public understanding can persist even when individual actors are rational and sincere. Second, it suggests that improving information quality alone may be insufficient if the architecture of message flow precludes coherence. And third, it reframes risk communication not simply as a matter of message clarity or factual correctness, but as a structural design problem: coherence must be engineered, not assumed.

There are several promising directions for future research. One is the development of methods for empirically estimating sheaf structures from real-world data. This includes mapping the belief stalks held by different agents and institutions, and inferring the effective restriction maps shaped by trust, framing, or institutional filters. Integrating these tools with machine learning approaches could also allow for more sensitive detection of misinformation—not just through content classification, but by identifying patterns of global inconsistency in belief networks. The sheaf theory perspective's novelty lies not in introducing another diffusion model but in providing computable structural diagnostics—sheaf cohomology and sheaf-based centralities—that detect hidden inconsistency and quantify influence under varying communication geometries and messaging strategies (e.g., honest vs. strategically distorted). These diagnostics can be applied to empirical data, allowing validation against observed persistence of disagreement or disproportionate influence. Future research should also develop larger scale empirical tests, examine time-varying or adaptive restriction maps, and integrate behavioral data on credibility and trust. In sum, sheaf theory complements rather than replaces prior opinion-dynamics and risk-amplification approaches, supplying the mathematical structure needed to quantify how misinformation and miscommunication reshape risk-communication flows and resulting agent beliefs, both true and expressed.

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

We believe that the sheaf-theoretic perspective can serve both as a conceptual framework and as a hypothesis-generating, predictive model. Conceptually, it specifies structural coherence conditions—global sections and cohomology—that determine whether locally consistent communications compose into a globally consistent belief state. Operationally, it yields testable predictions such as (i) persistence of disagreement when first cohomology is nontrivial ($H^1 \neq 0$); (ii) influence shifts under deception, whereby lower honesty parameters in restriction maps increase deceptive agents' sheaf-Laplacian centrality; (iii) intervention effects of reweighting trust/provenance channels, which are predicted to reduce dim H^1 and move equilibria toward the truthful harmonic subspace; and (iv) forecastable consensus trajectories derived from the sheaf Laplacian allowing out-of-sample evaluation of time-to-consensus and equilibrium location in expressed coordinates. These propositions connect the qualitative insights of SARF to quantitative, falsifiable implications for observed communication networks.

A proposed use of novel mathematical formalisms in risk communication—and in risk analysis more generally—should invite two healthy skeptical challenges: Is the formalism merely a way to adorn old concepts with new names, or is it substantive and helpful, leading to new insights, new methods for solving old problems, or articulation and formulation of new and worthwhile problems whose solution can benefit practitioners? If a new mathematical formalism is genuinely helpful, can it be expressed succinctly, with mathematical precision and elegance, uncluttered with heuristic motivations and imprecise interpretations and discussions?

In answer to the first, we think that the qualitative observation that different actors in a risk-communication system typically possess fragmented, at best only partially overlapping, context-dependent information and beliefs is not new (Aven and Thekdi 2022), but a sheaf-theoretic perspective provides a constructive framework for asking—and answering—questions that cannot easily be addressed with existing descriptive models. It allows intuitive ideas such as "partial overlap" to be defined more precisely (stalks represent the precise spaces of beliefs available to each actor; restriction maps encode how those beliefs must align when communication occurs) and then provides cohomology tests to identify when a collection of locally consistent messages fails to extend to any globally consistent interpretation. We believe that this goes beyond traditional probabilistic, behavioral, or network models that focus on *content accuracy*, *cognitive biases*, or *channels of transmission* by emphasizing structural incoherencies—obstructions to consistency—that persist even when all messages are accurate, honest, and mutually trusted.

The sheaf-Laplacian framework also offers new dynamic and predictive capabilities. Sheaf Laplacians generalize classical diffusion models by allowing information to propagate in ways that respect the agents' semantic constraints. This makes it possible to model how misaligned meanings evolve over time, to simulate interventions, and to identify stable belief patterns (such as persistent disagreement) that are structurally forced. These tools parallel well-established network-Laplacian methods, making them computationally tractable for large systems.

Perhaps most importantly, we believe that this framework opens new classes of questions for risk analysts—questions that are difficult to pose and analyze in previous models. These include:

- *Does a globally coherent shared understanding exist for this communication architecture?*
- *If not, which semantic maps or communication links must be modified to restore coherence?*
- *Where exactly does a structural obstruction arise, and how sensitive is it to alternative definitions or trust weights?*

These are questions of coherence architecture, not content accuracy, and existing methods cannot answer them. Sheaf theory provides the mathematical machinery to do so.

Applications in public health, environmental risk, and emergency response are particularly compelling. Systems designed for "cognitive interoperability"—that is, for the faithful and consistent transmission of risk-relevant beliefs across diverse actors—could be explicitly evaluated and improved using sheaf-theoretic diagnostics. There is also potential for fruitful integration with causal inference methods, especially those that rely on graphical models or influence diagrams. These could help clarify how misinformation disrupts not just shared beliefs but shared understandings of causal structure and responsibility.

More broadly, this work points toward a normative theory of design and evaluation for trustworthy information systems. In addition to concerns about accuracy and bias, such systems should be evaluated for their ability to support coherence, reconstructability, and epistemic resilience. Sheaf theory, despite its abstract origins, offers a possible foundation for articulating and achieving these goals.

In sum, this paper has proposed that many of the most persistent and dangerous failures in risk communication arise not from bad data or bad actors, but from structural misalignment in the architecture of belief exchange. By bringing the tools of sheaf theory to bear on the challenges of misinformation, we hope to provide both new explanatory power and practical methods for designing and operating more resilient, coherent systems of risk understanding and communication—systems that can support more effective distributed and decentralized decision-making in real-world risk management applications.

Acknowledgments

I used OpenAI's ChatGPT throughout the preparation of this paper in performing data analyses for the examples, generating code for the figures, and drafting initial interpretations of results and successive iterations of outlines and text. The author is responsible for all aspects of the final text. I believe that such human–AI collaborations highlight the potential of AI to help increase productivity and advance the quality of research. I thank three anonymous reviewers and Kash Barker for helpful comments and Jeff Lewis for a stimulating discussion that inspired this research.

Funding

The author has nothing to report.

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Conflicts of Interest

The author declares no conflicts of interest.

References

Acemoglu, D., A. Ozdaglar, and A. ParandehGheibi. 2010. "Spread of (Mis)Information in Social Networks." *Games and Economic Behavior* 70, no. 2: 194–227. https://doi.org/10.1016/j.geb.2010.01.005.

Aktas, M. E., E. Akbas, and A. Hahn. 2024. "Liars Are More Influential: Effect of Deception in Influence Maximization on Social Networks." In *2024 IEEE International Conference on Big Data (BigData); 2024 Dec 15–18; Washington, DC, USA*, 10938987. Piscataway (NJ): IEEE. https://doi.org/10.1109/BigData62323.2024.10938987. https://arxiv.org/abs/2203.11275.

Allassan, F. 2020. "Surgeon General Defends Reversal on Face Mask Policy." *Axios [Internet]*, July 12. Accessed May 24, 2025. https://www.axios.com/2020/07/12/surgeon-general-reversal-face-mask.

Aven, T., and S. A. Thekdi. 2022. "On How to Characterize and Confront Misinformation in a Risk Context." *Journal of Risk Research* 25, no. 11–12: 1272–1287.

Banerjee, A. V. 1992. "A Simple Model of Herd Behavior." *Quarterly Journal of Economics* 107, no. 3: 797–817.

Bautista, A. 2025 Jan. "Opinion Dynamics in Bounded Confidence Models With Manipulative Agents: Moving the Overton Window." *Physica A* 660: 130379. https://doi.org/10.1016/j.physa.2025.130379.

Bikhchandani, S., D. Hirshleifer, and I. Welch. 1992. "A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades." *Journal of Political Economy* 100, no. 5: 992–1026.

Brewer, D. D. 2000. "Forgetting of Friends and Its Effects on Measuring Friendship Network Characteristics and Properties." *Social Networks* 22, no. 1: 29–43.

Butts, D. J., S. A. Bollman, and M. S. Murillo. 2023. "Mathematical Modeling of Disinformation and Effectiveness of Mitigation Policies." *Scientific Reports* 13: 18735. https://doi.org/10.1038/s41598-023-45710-2.

Coleman, J., E. Katz, and H. Menzel. 1957. "The Diffusion of an Innovation Among Physicians." *Sociometry* 20, no. 4: 253.

Coleman, J. S. 1964. *Introduction to Mathematical Sociology*. Free Press.

Cox, L. A. Jr. 2019 Nov. "Improving Causal Determination." *Global Epidemiology* 1: 100004. https://doi.org/10.1016/j.gloepi.2019.100004.

Curry, J. 2013. *Sheaves, Cosheaves and Applications*. arXiv preprint arXiv:1303.3255 [math.AT]. 188. https://arxiv.org/abs/1303.3255.

Deffuant, G., D. Neau, F. Amblard, and G. Weisbuch. 2000. "Mixing Beliefs Among Interacting Agents." *Advances in Complex Systems* 03, no. 1–4: 87–98. https://doi.org/10.1142/S0219525900000078.

DeGroot, M. H. 1974. "Reaching a Consensus." *Journal of the American Statistical Association* 69, no. 345: 118–121. https://doi.org/10.2307/2285509.

Del Vicario, M., A. Scala, G. Caldarelli, H. E. Stanley, and W. Quattrociocchi. 2017. "Modeling Confirmation Bias and Polarization." *Scientific Report* 7: 40391. https://doi.org/10.1038/srep40391.

Erokhin, D. 2025. "Public Discourse Surrounding the 2025 California Wildfires: A Sentiment and Topic Analysis of High-Engagement YouTube Comments." *Geosciences* 15, no. 3: 100. https://doi.org/10.3390/geosciences15030100.

Fowler, J. H. 2006. "Legislative Co-Sponsorship Networks in the U.S. House and Senate." *Social Networks* 28, no. 4: 454–465.

Friedkin, N. E., and E. C. Johnsen. 1990. "Social Influence and Opinions." *Journal of Mathematical Sociology* 15, no. 3–4: 193–206. https://doi.org/10.1080/0022250X.1990.9990069.

Friedkin, N. E., and E. C. Johnsen. 1999. "Social Influence Networks and Opinion Change." *Advances in Group Processes* 16: 1–29.

Gleiser, P., and L. Danon. 2003. "Community Structure in Jazz." *Advances in Complex Systems* 06, no. 4: 565–573.

Golub, B., and M. O. Jackson. 2010. "Naïve Learning in Social Networks and the Wisdom of Crowds." *American Economic Journal Microeconomics* 2, no. 1: 112–149. https://doi.org/10.1257/mic.2.1.112.

Hansen, J., and R. Ghrist. 2019. "Toward a Spectral Theory of Cellular Sheaves." *Journal of Applied and Computational Topology* 3, no. 3–4: 315–358. https://doi.org/10.1007/s41468-019-00038-7.

Hansen, J., and R. Ghrist. 2021. "Opinion Dynamics on Discourse Sheaves." *Siam Journal on Applied Mathematics* 81, no. 5: 2033–2060. https://doi.org/10.1137/20M1341088.

Hegselmann, R., and U. Krause. 2002. "Opinion Dynamics and Bounded Confidence: Models, Analysis, and Simulation." *Journal of Artificial Societies and Social Simulation* 5, no. 3. http://jasss.soc.surrey.ac.uk/5/3/2.html.

Jamalzadeh, S., K. Barker, A. D. González, and S. Radhakrishnan. 2022. "Protecting Infrastructure Performance From Disinformation Attacks." *Scientific Reports* 12, no. 1: 12707. https://doi.org/10.1038/s41598-022-16832-w. PMID: 35882902; PMCID: PMC9325778.

Jeong, D., and E. Shin. 2024. "Optimal Influence Design in Networks." *Journal of Economic Theory* 220: 105877. https://doi.org/10.1016/j.jet.2024.105877.

Kasperson, R. E., T. Webler, B. Ram, and J. Sutton. 2022. "The Social Amplification of Risk Framework: New Perspectives." *Risk Analysis* 42, no. 7: 1367–1380. https://doi.org/10.1111/risa.13926 PMID: 35861634; PMCID: PMC10360138.

Kasperson, R. E., O. Renn, P. Slovic, et al. 1988. "The social amplification of risk: A conceptual framework." *Risk Analysis* 8, no. 2: 177–188. https://doi.org/10.1111/j.1539-6924.1988.tb01168.x.

Knuth, D. E. 1993. *The Stanford GraphBase: A Platform for Combinatorial Computing*. Addison-Wesley.

Koutis, I., G. L. Miller, and R. Peng. 2011. "A Nearly-m Log n Time Solver for SDD Linear systems." In *2011 IEEE 52nd Annual Symposium on Foundations of Computer Science (FOCS). Oct 22–25; Palm Springs, CA*, 590–598. IEEE. https://doi.org/10.1109/FOCS.2011.85.

McCandless Farmer, B. 2020. "Anthony Fauci Talks With Dr. Jon LaPook About COVID-19." 60 Minutes Overtime [Internet]. *CBS News*, Mar 8. Accessed May 24, 2025. https://www.cbsnews.com/news/preventing-coronavirus-facemask-60-minutes-2020-03-08/.

Milli, L. 2021. "Opinion dynamic modeling of news perception." *Applied Network Science* 6, no. 1: 76. https://doi.org/10.1007/s41109-021-00412-4.

Moussaïd, M., H. Brighton, and W. Gaissmaier. 2015. "The Amplification of Risk in Experimental Diffusion Chains." *PNAS* 112, no. 18: 5631–5636. https://doi.org/10.1073/pnas.1421883112.

Murdock, I., K. M. Carley, and O. Yağan. 2024. "An Agent-Based Model of Cross-Platform Information Diffusion and Moderation." *Social Network Analysis and Mining* 14, no. 145. https://doi.org/10.1007/s13278-024-01305-x.

Newman, M. E. J. 2006. "Finding Community Structure in Networks Using the Eigenvectors of Matrices." *Physical Review E* 74, no. 3: 036104.

Pitigraisorn, P. 2020."Disinformation, Science Communication and Trust: Food Rumours in Thailand." PhD diss., University of Essex. https://repository.essex.ac.uk/29825/.

Rodríguez, J. A. 2004. *Madrid Train Bombing Network [Internet]*. University of Barcelona. Accessed Oct 4, 2025. http://users.dimi.uniud.it/~massimo.franceschet/ns/syllabus/make/madrid/madrid.html.

Rosiak, D. 2022. *Sheaf Theory Through Examples*. MIT Press. https://doi.org/10.7551/mitpress/12581.003.0012.

Spielman, D. A., and S. H. Teng. 2011. "Spectral Sparsification of Graphs." *Siam Journal on Computing* 40, no. 4: 981–1025. https://doi.org/10.1137/08074489X.

15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Tambuscio, M., and G. Ruffo. 2019. "Fact-Checking Strategies to Limit Urban Legends Spreading in a Segregated Society." *Applied Network Science* 4: 116. https://doi.org/10.1007/s41109-019-0233-1.

Wu, Y., and P. Guo. 2024. "Modeling Misinformation Spread in a Bounded Confidence Model." *Entropy* 26, no. 2: 99. https://doi.org/10.3390/e26020130.

Appendix A: R code for models

```
# R code for Figure 1

# DeGroot model: 5 agents, including a persistent activist

# Define the weight matrix W (row-stochastic)

W ← matrix(c(

  0.60, 0.20, 0.10, 0.05, 0.05, # Regulator
  0.30, 0.50, 0.10, 0.05, 0.05, # Corporation
  0.20, 0.20, 0.50, 0.10, 0.00, # Citizen A
  0.10, 0.20, 0.10, 0.50, 0.10, # Citizen B
  0.00, 0.00, 0.00, 0.00, 1.00 # Activist

), nrow = 5, byrow = TRUE)

# Initial beliefs

  beliefs ← c(0.2, 0.25, 0.2, 0.3, 1.0)

# Number of time steps

  n_steps ← 50

# Store belief trajectories

  belief_history ← matrix(0, nrow = n_steps, ncol = length(beliefs))

  belief_history[1,] ← beliefs

# Simulate updates

  for (t in 2:n_steps) {

  belief_history[t,] ← W %*% belief_history[t - 1,]

}

# Plot with thicker lines

  matplot(belief_history, type = "l", lty = 1, col = 1:5, lwd = 2,

  xlab = "Time", ylab = "Belief",

  main = "DeGroot Opinion Dynamics (50 Time Steps)")

  legend("bottomright", legend = c("Regulator", "Corporation", "Citizen A", "Citizen B", "Activist"),

  col = 1:5, lty = 1, lwd = 2)
```

 # Friedkin–Johnsen model: 5 agents, including a persistent activist

```
# R code for Figure 2

# Define the weight matrix W (row-stochastic)

W ← matrix(c(

  0.60, 0.20, 0.10, 0.05, 0.05, # Regulator
  0.30, 0.50, 0.10, 0.05, 0.05, # Corporation
  0.20, 0.20, 0.50, 0.10, 0.00, # Citizen A
  0.10, 0.20, 0.10, 0.50, 0.10, # Citizen B
  0.00, 0.00, 0.00, 0.00, 1.00 # Activist (fully stubborn)

), nrow = 5, byrow = TRUE)

# Initial beliefs

  x0 ← c(0.2, 0.25, 0.2, 0.3, 1.0)

# Anchoring strengths (0 = fully open to influence, 1 = completely stubborn)

  alpha ← c(0.1, 0.2, 0.3, 0.2, 1.0) # activist is fully anchored; others vary

A ← diag(alpha) # anchoring matrix

I ← diag(5) # identity matrix

# Number of time steps

  n_steps ← 50

# Store belief trajectories

  belief_history ← matrix(0, nrow = n_steps, ncol = length(x0))

  belief_history[1,] ← x0

# Simulate Friedkin–Johnsen updates

  for (t in 2:n_steps) {

  belief_history[t,] ← A %*% x0 + (I—A) %*% W %*% belief_history[t - 1,]

}

# Plot belief trajectories over time

  matplot(belief_history, type = "l", lty = 1, col = 1:5, lwd = 2,

  xlab = "Time", ylab = "Belief",

  main = "Friedkin–Johnsen Opinion Dynamics (Anchored Beliefs)")

# Move legend to top left to avoid overlap with curves

  legend("topleft", legend = c("Regulator", "Corporation", "Citizen A", "Citizen B", "Activist"),

  col = 1:5, lty = 1, lwd = 2, bty = "n")
```

 # Hegselmann–Krause (HK) bounded confidence model in R

```
# R code for Figure 3
```


15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

```
# Parameters
  n_agents ← 10 # number of agents
  epsilon ← 0.2 # bounded confidence threshold
  n_steps ← 10 # number of time steps
# Initial beliefs: random values in [0, 1]
  set.seed(123)
  beliefs ← runif(n_agents)
# Store belief history
  belief_history ← matrix(0, nrow = n_steps, ncol = n_agents)
  belief_history[1,] ← beliefs
# HK update function: average beliefs within epsilon range
  hk_update ← function(x, eps) {
  updated ← numeric(length(x))
  for (i in 1:length(x)) {
  neighbors ← which(abs(x—x[i]) ≤ eps)
  updated[i] ← mean(x[neighbors])
}
  return(updated)
}
# Simulate updates
  for (t in 2:n_steps) {
  beliefs ← hk_update(beliefs, epsilon)
  belief_history[t,] ← beliefs
}
# Plot
  matplot(belief_history, type = "l", lty = 1, lwd = 2,
  col = rainbow(n_agents), xlab = "Time", ylab = "Belief",
  main = paste("Hegselmann–Krause Dynamics (epsilon = ", epsilon, ")"))
  legend("right", legend = paste("Agent", 1:n_agents), col = rainbow(n_agents), lty = 1, lwd = 2, cex = 0.7, bty = "n")
```

 # Deffuant–Weisbuch (DW) bounded confidence model in R

```
# R code for Figure 4
# Parameters
  n_agents ← 10 # number of agents
  epsilon ← 0.2 # confidence threshold
  mu ← 0.5 # convergence parameter
  n_steps ← 300 # number of pairwise interactions
# Initial beliefs: random values in [0, 1]
  set.seed(456)
  beliefs ← runif(n_agents)
# Store belief history
  belief_history ← matrix(NA, nrow = n_steps + 1, ncol = n_agents)
  belief_history[1,] ← beliefs
# DW update: pairwise interactions
  for (t in 1:n_steps) {
  pair ← sample(1:n_agents, 2, replace = FALSE)
  i ← pair[1]; j ← pair[2]
  diff ← abs(beliefs[i]—beliefs[j])
  if (diff ≤ epsilon) {
  avg ← (beliefs[i] + beliefs[j]) / 2
  beliefs[i] ← beliefs[i] + mu * (beliefs[j]—beliefs[i])
  beliefs[j] ← beliefs[j] + mu * (beliefs[i]—beliefs[j])
}
  belief_history[t + 1,] ← beliefs
}
# Plot belief trajectories
  matplot(belief_history, type = "l", lty = 1, lwd = 2,
  col = rainbow(n_agents), xlab = "Interaction Step",
  ylab = "Belief", main = paste("Deffuant–Weisbuch Dynamics (epsilon = ", epsilon, ", mu = ", mu, ")"))
  legend("right", legend = paste("Agent", 1:n_agents),
  col = rainbow(n_agents), lty = 1, lwd = 2, cex = 0.7, bty = "n")
```

 # Hegselmann–Krause bounded confidence model augmented with a trusted bridge communicator

```
# R code for Figure 5
# Parameters
  n_agents ← 10
```


15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

```
epsilon ← 0.2

n_steps ← 10

trusted_agent ← 3 # Agent index (1-based in R) who will be trusted by all others

# Polarized initial beliefs: 3 agents near 0.2, 4 near 0.5, 3 near 0.8

initial_beliefs ← c(0.2, 0.22, 0.18, 0.5, 0.48, 0.52, 0.51, 0.8, 0.78, 0.82)

# Update function for standard HK model

hk_update ← function(x, eps) {

updated ← numeric(length(x))

for (i in 1:length(x)) {

neighbors ← which(abs(x—x[i]) ≤ eps)

updated[i] ← mean(x[neighbors])

}

return(updated)

}

# Update function with trusted agent

hk_update_trusted ← function(x, eps, trusted) {

updated ← numeric(length(x))

for (i in 1:length(x)) {

neighbors ← which(abs(x—x[i]) ≤ eps)

if (i ! = trusted && !(trusted %in% neighbors)) {

neighbors ← c(neighbors, trusted)

}

updated[i] ← mean(x[neighbors])

}

return(updated)

}

# Run simulation and store history

simulate_hk ← function(update_func, . . . ) {

beliefs ← initial_beliefs

history ← matrix(NA, nrow = n_steps, ncol = n_agents)

history[1,] ← beliefs

for (t in 2:n_steps) {

beliefs ← update_func(beliefs, epsilon, . . . )

history[t,] ← beliefs

}

return(history)

}

# Run both scenarios

history_standard ← simulate_hk(hk_update)

history_trusted ← simulate_hk(hk_update_trusted, trusted = trusted_agent)

# Plot: two-panel comparison

op ← par(mfrow = c(1, 2), mar = c(4, 4, 3, 1)) # Save original graphics settings

# Panel (i): No trusted agent

matplot(history_standard, type = “l”, lty = 1, lwd = 2,

col = rainbow(n_agents), xlab = “Time”, ylab = “Belief”,

main = “No Trusted Bridging Agent”)

legend(“topright”, legend = paste(“Agent”, 1:n_agents), col = rainbow(n_agents),

lty = 1, lwd = 2, cex = 0.6, bty = “n”)

# Panel (ii): One trusted bridging agent

matplot(history_trusted, type = “l”, lty = 1, lwd = 2,

col = rainbow(n_agents), xlab = “Time”, ylab = “Belief”,

main = paste(“Trusted Bridging Agent:”, paste(“Agent”, trusted_agent)))

legend(“topright”, legend = paste(“Agent”, 1:n_agents), col = rainbow(n_agents),

lty = 1, lwd = 2, cex = 0.6, bty = “n”)

# Reset graphics parameters

par(op)
```

Appendix B: R Code for Deceptive Communication Example

```
## —- Four-node sheaf-Laplacian toy example ————————————————

## Coboundary operator for the line graph 1—2—3—4

D0 ← matrix(c(

1,-1, 0, 0,
0, 1,-1, 0,
```


15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

```
  0, 0, 1,-1

), nrow = 3, byrow = TRUE)

## Sheaf Laplacian

L ← t(D0) %*% D0

## Helper: Laplacian "energy" per node and normalized centrality

  lap_centrality ← function(x, L) {

Lx ← as.vector(L %*% x)

  e ← Lx^2

  c ← e / sum(e)

  list(Lx = Lx, energy = e, centrality = c)

}

## ————————- Honest case ————————-

x_honest ← c(1.0, 0.6, -0.2, -1.0)

  res_honest ← lap_centrality(x_honest, L)

  cat("\nHONEST CASE\n")

  cat("x = ", paste(round(x_honest, 3), collapse = " "), "\n")

  cat("D0 x = ", paste(round(as.vector(D0 %*% x_honest), 3), collapse = " "), "\n")

  cat("L x = ", paste(round(res_honest$Lx, 3), collapse = " "), "\n")

  cat("energy = ", paste(round(res_honest$energy, 3), collapse = " "), "\n")

cat("centrality = ", paste(round(res_honest$centrality, 6), collapse = " "), "\n")

## ————————- Deceptive case ————————-

## Honesty level h = 0.5 for nodes 2 and 3, with bias b2 = +1.0, b3 = -1.0.

  h ← 0.5

  b ← c(NA, 1.0, -1.0, NA) # NA means "no change" for nodes 1 and 4

  x_exp ← x_honest

  x_exp[2] ← h * x_honest[2] + (1 - h) * b[2]

  x_exp[3] ← h * x_honest[3] + (1 - h) * b[3]

## nodes 1 and 4 remain honest: r_i(x_i) = x_i

  res_deceptive ← lap_centrality(x_exp, L)

  cat("\nDECEPTIVE CASE (h = 0.5 for nodes 2 & 3; b2 = +1.0, b3 = -1.0)\n")

  cat("x_exp = ", paste(round(x_exp, 3), collapse = " "), "\n")

  cat("D0 x_exp = ", paste(round(as.vector(D0 %*% x_exp), 3), collapse = " "), "\n")

  cat("L x_exp = ", paste(round(res_deceptive$Lx, 3), collapse = " "), "\n")

  cat("energy_exp = ", paste(round(res_deceptive$energy, 3), collapse = " "), "\n")

  cat("centrality_exp = ", paste(round(res_deceptive$centrality, 6), collapse = " "), "\n")

## Optional: quick barplots to visualize the shift

  op ← par(mfrow = c(1,2), mar = c(4,4,2,1))

  barplot(res_honest$centrality, names.arg = 1:4,

  main = "Honest centrality", ylab = "Normalized", ylim = c(0,1))

  barplot(res_deceptive$centrality, names.arg = 1:4,

  main = "Deceptive centrality", ylab = "Normalized", ylim = c(0,1))

par(op)
```


15396924, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/risa.70224, Wiley Online Library on [30/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License